Structural Reliability Under Projection: Representational Loss, Reconstructibility, and Admissibility

Brian Cameron

September 6, 2026

Abstract

AI systems reason and act from compressed representations such as prompts, summaries, retrieved passages, logs, metrics, memory, and policy artifacts. Such representations may collapse distinctions required by downstream claims while leaving those claims locally coherent. This paper characterizes that failure as query-relative representational loss rather than ordinary inference error. For a projection $P : S \to R$, a query is identifiable only when states collapsed by P do not require different answers.

The paper introduces dependency status as a thin operational interface between projected representation and epistemic operation. Preserved, reconstructible, traceable, and opaque status arise from a gated recovery–attribution construction. Under explicit surjectivity, status-sufficiency, and fiber-separation assumptions, these four statuses are exactly the behavioral equivalence classes induced by a declared base licensing relation; three nested capability thresholds make the partition explicit, and the existence of operations witnessing those thresholds provides a concrete sufficient condition for fiber separation. Across a status-separating family of contexts, no proper quotient preserves the same portable licensing behavior.

Dependency status constrains epistemic licensing; operational validity separately constrains permission; and a compliant selection policy prescribes from their intersection while actual selection remains observed behavior. This separation distinguishes representation insufficiency, assessment failure, reporting failure, and action-governance failure, while unsupported promotion names the more general pattern in which a downstream role or operation acquires stronger support than the representation supplies. The resulting framework treats dependency status as a bounded cybernetic control interface: changes in represented support should change the operation or response form prescribed for the reasoner without implying perfect diagnosis, behavioral compliance, convergence, or self-certification.

1 Introduction: From Representational Support to Licensed Operation

AI systems increasingly reason and act from compressed representations: prompts, summaries, retrieved passages, logs, traces, metrics, memories, intermediate reasoning states, and policy artifacts. Such representations may support fluent and locally coherent reasoning while failing to preserve a distinction required by the resulting claim or action. The system can therefore produce an answer that looks well reasoned even though its operative information basis does not determine the conclusion it presents.

This paper develops a conceptual bridge between formal accounts of identifiability and practical concerns in AI reasoning, safety, security, and agentic reliability. Its novelty claim is not that projection can destroy identifiability, sufficiency, or task-relevant information. Those phenomena are well established in statistics, decision theory, abstraction, partial-observation models, and information theory. The additional step taken here is operational: the relation between a claim and its represented support constrains which epistemic operation the reasoner may perform and the form in which its result may be communicated.

The architecture can be summarized as

$$\boxed{\text{projection} \to \text{dependency status} \to \text{licensed epistemic operation} \to \text{response or action.}}$$

A projected representation may preserve a required distinction directly, make it recoverable through a bounded procedure, retain only an accountable trace to unavailable support, or fail to preserve even an adequate dependency trail. These conditions are represented by the dependency statuses preserved, reconstructible, traceable, and opaque. Dependency status functions as a thin interface between heterogeneous representations and heterogeneous reasoning operations: prompts, telemetry, retrieved sources, metrics, causal models, memories, and policy artifacts may all be assessed through the same operational question of what happened to the dependency required by a proposed continuation.

The downstream operations are likewise heterogeneous. Depending on represented support, a reasoner may infer directly, reconstruct, retrieve, inspect, qualify, branch over unresolved alternatives, reframe the question, route the problem, act under an explicit decision rule, defer, or refuse the requested form. The central thesis is therefore that structural insufficiency should change the available epistemic operation, not merely reduce confidence while leaving the operation unchanged.

Dependency status does not by itself prescribe one continuation or compel a system to follow it. It constrains the epistemically licensed operation set. Operational validity then determines which licensed continuations are admissible. Selection and budget policy may prescribe among admissible continuations according to cost, expected diagnostic value, utility, urgency, policy, reversibility, and other represented decision commitments. The continuation actually selected remains observed behavior and may comply with or depart from that prescription. Accordingly,

$$\boxed{\text{structural support} \neq \text{operational permission} \neq \text{policy prescription} \neq \text{actual selection.}}$$

The framework also distinguishes the dependency status that actually obtains from the status a bounded reasoner assesses, the status it reports, the operation prescribed by a compliant policy, and the operation actually selected. A system can follow its operation policy faithfully while having incorrectly assessed the adequacy of its representation. It can assess its support correctly while reporting a stronger status, or it can report its assessment faithfully while selecting an operation that violates the assessed admissibility constraints or policy prescription. Factual correctness, assessment fidelity, reporting fidelity, prescription fidelity, and governance compliance are therefore related but non-equivalent properties.

The shared structural failure is *unsupported promotion*: a downstream object or operation acquires a stronger epistemic role than its represented support warrants. A traceable source pointer is treated as reconstructed evidence; a policy-selected action is reported as evidence-determined; a proxy is treated as its target; a locally supported judgment is transferred to a composed system; or an executable procedure is treated as licensed merely because it remains available. These failures differ at the domain level, but they share the same abstract form:

$$\boxed{\text{weaker or differently typed support} \not\Rightarrow \text{stronger epistemic license.}}$$

The objective is not representational completeness or the elimination of all error. A bounded system cannot guarantee that it has discovered every material dependency or correctly assessed every dependency it represents. The more attainable objective is diagnostic legibility: consequential insufficiency, unresolved alternatives, unsupported assumptions, and points of loss should remain visible enough to constrain response form and, where practical, motivate bounded correction or targeted representation refinement.

This produces a cybernetic interpretation of reasoning under projection. A bounded reasoner operates from a representation, assesses the status of dependencies required by a proposed continuation, restricts its available operations, observes the consequences of those operations, and may refine its representation when recurring failure demonstrates that an omitted distinction matters. This is a corrigibility mechanism rather than a guarantee of convergence or self-certification.

Contributions. The paper makes five connected contributions:

1. It distinguishes correctable inference error from query-relative representational loss and states the resulting non-identifiability boundary formally.

2. It derives preserved, reconstructible, traceable, and opaque status from a gated recovery–attribution construction, expresses that construction through three nested capability thresholds, and proves under stated sufficiency, separation, and surjectivity assumptions when the statuses are exactly the primitive behavioral classes.

3. It characterizes dependency status as a thin licensing interface, distinguishes controller-local compression from portable status non-collapse, and separates epistemic licensing, operational validity, admissibility, policy prescription, and actual selection.

4. It identifies unsupported promotion as a common structural failure and separates actual, assessed, and reported dependency status from the policy-prescribed operation and the operation actually selected.

5. It develops observability, bounded correction, and non-collapse as requirements for keeping dependency-status governance effective in a corrigible but non-self-certifying control loop.

The paper first introduces a minimal case in which projection makes a query non-identifiable. It then develops projection, error, loss, and dependency status; connects status to licensed epistemic operation; and derives the structural requirements for reliable correction. Diagnostic cases show how the same interface applies to scalarization, causal attribution, provenance, decision under non-identifiability, and agentic action. The final sections relate the framework to neighboring methods, instantiate it as a five-step operational diagnostic, and describe its cybernetic implications and limitations.

2 A Minimal Motivating Example

Let the underlying state space contain two states,

$$S = \{s_1, s_2\},$$

and suppose that projection maps both states to the same representation:

$$P(s_1) = P(s_2) = r.$$

For a query $q : S \to A$, suppose instead that

$$q(s_1) \neq q(s_2).$$

The reasoner receives the same operative representation r in both cases, even though the query requires different answers. No function operating only on r can recover q correctly for both underlying states. A determinate answer therefore requires additional information, representation expansion, an explicit assumption or decision rule, or a change in answer form.

This failure differs from ordinary inference error. If the reasoner emits $q(s_1)$ and the underlying state happens to be s_1, the answer is factually correct, but the epistemic operation remains unsupported by the representation. The same operation would have produced the wrong answer under s_2, and nothing in r distinguishes the two cases. Realized accuracy does not retroactively provide the missing structural support.

Consider an AI assistant operating during a security incident. It receives a compressed summary of logs, deployment history, alerts, and recent configuration changes. The summary may be compatible with a deploy-triggered bug, a dependency outage, a configuration interaction, or telemetry loss caused by reduced logging. If the representation does not distinguish those states, naming one as the root cause is not merely a risky inference. It is a determinate attribution from an information basis that does not support one.

The appropriate continuation need not be blanket abstention. The assistant may request additional telemetry, retrieve deployment or dependency records, branch over the remaining explanations, qualify the diagnosis, recommend a reversible diagnostic intervention, or act under an explicit incident policy that remains valid across the unresolved states. What it may not do is present one explanation as determined by the represented evidence when the required distinction has been projected away.

This example contains the paper's complete operational problem in minimal form. Projection determines what distinctions survive; dependency status constrains which epistemic operations are licensed; operational validity determines which licensed continuations are admissible; and separate budget or selection policies choose among the admissible continuations.

3 Reasoning Under Projection

To understand these failures, we must examine the conditions under which reasoning operates. In practice, reasoning systems do not operate on complete representations of the problems they address.

3.1 Reasoning Over Representations

A reasoning process operates on a representation of a problem rather than on the full underlying system. In an AI system, that representation may include a prompt, retrieved context, tool output, exposed memory, log summary, schema, score, or policy artifact. The relevant question is not whether the representation is complete, but whether it preserves the distinctions required by the proposed claim or continuation.

Operative representation in LLM systems. The operative representation is not necessarily identical to the user prompt. It consists of the information basis available for the current reasoning step, including explicitly available conversation state, retrieval, tool output, background knowledge, assumptions, and policy constraints. Persistent information enters that basis only when it is retrieved, exposed, or otherwise made available through a permitted bounded procedure. Thus,

$$\boxed{\text{persistent represented state} \neq \text{current operative representation.}}$$

The boundary need not be perfectly inspectable for the framework to apply. The requirement is narrower: when an answer materially depends on background knowledge, an external source, an assumption, or a tool result, that support role should not be presented as though it were determined by visible evidence alone.

Where these roles matter, write the operative representation schematically as

$$R = \langle R_E, R_B, R_D \rangle,$$

where R_E contains represented evidence or observations, R_B contains invoked background knowledge, priors, assumptions, or model commitments, and R_D contains policy, utility, authority, or other operational constraints. These components need not be independent or separately implemented. The notation preserves their different licensing roles:

$$\boxed{\text{evidence-supported} \neq \text{prior- or assumption-supported} \neq \text{policy-selected.}}$$

Outcome-determining provenance and authority are themselves typed, query-relative dependencies. Preserving a source does not necessarily preserve whether it merely supported, challenged, triggered, or selected an outcome [12].

3.2 Projection as Compression

A *projection* is a transformation from a richer state into the reduced representation used for reasoning:

$$P : S \to R.$$

Although projection may be lossy in the information-theoretic sense [24], the present concern is task-relative: whether R preserves the distinctions required by a particular inference, decision, or action. A highly compressed representation may be sufficient for one query and inadequate for another.

The richer state S need not be an inaccessible world state. It may be a persistent reasoning artifact from which a bounded operative context is assembled:

$$P_q : B \to R_q.$$

Keeping B available can make later recovery more explicit and repeatable, but it does not eliminate projection. Only the state actually available directly or through a permitted bounded recovery path supports the current continuation. Persistence is therefore neither permanence nor present support.

In many AI settings, S is itself a modeled or idealized state rather than a complete description of reality. The framework is consequently relative both to the selected state model and to the projection by which that model becomes operationally available.

3.3 Target Formation and Representational Projection

The mapping $P : S \to R$ begins from a state space S already treated as the target of reasoning. Swanson distinguishes an upstream boundary operation that stabilizes a target from a subsequent rendering operation that produces a finite representation of that target [28]. Adapting that distinction to the present notation, a richer account may therefore distinguish target formation from subsequent projection:

$$F \xrightarrow{B_\Gamma} S_\Gamma \xrightarrow{P_\Gamma} R,$$

where F denotes a broader field of potentially relevant state, B_Γ selects or constructs the state space treated as relevant under regime Γ, and P_Γ maps that selected state into the operative representation.

This distinction separates two structurally different sources of insufficiency. *Target-selection loss* occurs when B_Γ excludes a distinction that is material to the downstream query. *Representational loss* occurs when the selected state S_Γ contains the relevant distinction but P_Γ fails to preserve or boundedly reconstruct it in R. The simpler notation $P : S \to R$ used throughout this paper should therefore be understood as conditioning on a currently declared target state S; it does not imply that the construction of S was itself lossless or complete.

The distinction matters operationally because representation repair and target revision are not the same intervention. Enriching R may repair a lossy projection from S_Γ while leaving untouched a material distinction that was excluded when S_Γ was formed. Conversely, revising the target boundary may reveal that a previously adequate projection was answering the wrong or incomplete question.

This is a distinction between stages of transformation rather than between independent reasoners. The two mappings are separated because failure at B_Γ and failure at P_Γ require different diagnoses and different repairs. Their appearance as a two-stage diagram should therefore not be generalized into a claim that reconstructible reasoning is inherently organized into pairs.

Multiple distinct states in S may map to the same representation in R.

In information-theoretic terms, R can be treated as the information basis available to the reasoner after projection. The relevant question is not whether R contains less information than S in general, but whether it retains the information needed to answer the query at hand. A representation may be highly compressed and still sufficient for one query, while being insufficient for another query that depends on a collapsed distinction.

A simple example illustrates the query-relative nature of projection. A black-and-white image may preserve shape while losing color. It remains admissible for many shape queries, but not for a query that depends on whether an object was red or green. The projection is not defective in general; it is insufficient for queries whose answers depend on the collapsed distinction.

Projection may likewise preserve procedures rather than only observable features. A compressed representation may retain a rule such as "when condition x appears, apply operation f" while omitting distinctions that originally limited when f was appropriate. The procedure therefore remains executable even though the query "is f licensed in this case?" may no longer be identifiable from the compressed representation. This is an important form of representational loss because operational continuity can conceal contextual loss: the persistence of a usable transformation may create the appearance that the conditions supporting that transformation were preserved as well.

Equivalently, projection determines which distinctions remain observable through the representation. This observable distinction structure can also be expressed using the language of representation-induced σ-algebras: a query is directly supported only when the distinction it requires is measurable under the projected representation, or becomes available through bounded reconstruction.

In this paper, projection refers specifically to transformations that induce non-trivial equivalence classes over the underlying state space: distinct states in S are mapped to the same representation in R.

The analysis is relative to a given projection P. A query may fail to be identifiable under one representation while becoming identifiable under a richer or differently structured representation. For example, a log summary may preserve that an outage occurred while losing the event ordering needed for root-cause analysis. A retrieval result may preserve a citation identifier while omitting the source passage needed to verify the claim. A risk score may preserve a ranking while losing the distinction between reversible and irreversible failures. These projections are useful, but they are not equally admissible for every downstream query.

In practice, however, reasoning systems often operate under fixed or externally imposed projections: a prompt, summary, metric, ranking, category, analogy, translation, dataset schema, interface boundary, token budget, or institutional decision format may determine what information is available at inference time. The question addressed here is therefore not whether some better representation could exist, but what follows when reasoning must proceed under the representation actually available.

3.4 Equivalence Classes and Collapse of Distinction

Because projection is many-to-one, it induces equivalence classes over the original state space. Distinct underlying states become indistinguishable once projected into the representation.

This means that some distinctions present in the original system may no longer be available to the reasoning process through the current representation. For the query being asked, those distinctions are not merely hidden within the representation; they are unavailable unless additional assumptions, structure, or observations are introduced.

This is the point at which ordinary partial observability and representational loss diverge. In a partial-observation setting, the system may maintain a belief state over hidden possibilities still represented by the model. Under projection-induced loss, the distinction required by the query may not be represented in the current state space at all. No update over the current representation can recover such a distinction unless the representation is expanded, a bounded reconstruction path is available, or an explicit assumption is introduced and marked.

3.5 Identifiability Under Projection

Let $P : S \to R$ be a projection from system states to representations. The projection induces an equivalence relation on S:

$$s_1 \sim_P s_2 \quad \text{iff} \quad P(s_1) = P(s_2).$$

A query, judgment, or decision-relevant property $q : S \to A$ is identifiable under P when it is constant over the equivalence classes induced by P:

$$P(s_1) = P(s_2) \Rightarrow q(s_1) = q(s_2).$$

For normative or evaluative queries, q should be read as model-relative: the framework analyzes whether a declared evaluative target is preserved under projection, not whether the target has a mind-independent unique value.

The use of identifiability here generalizes the term beyond parameter recovery in statistical or causal models. In this paper, identifiability refers to whether any reasoning-relevant query, judgment, or decision property survives projection into the representation used by the system.

If this condition holds, then the representation preserves enough information to answer q. If the condition fails, then two states that are indistinguishable in the representation require different answers to q. In that case, no reasoning process operating only on R can determine q without introducing additional assumptions, expanding the representation, or obtaining new information.

This gives a compact way to state the error–loss distinction. In the measure-theoretic formulation developed later, this corresponds to whether the query is measurable with respect to the observable structure induced by the projection. An error occurs when q is identifiable under the projection but the reasoning process returns the wrong value. Loss occurs when q is not identifiable under the projection, so the representation itself no longer supports a determinate answer.

In more familiar statistical and information-theoretic terms, error is a recoverable inference or estimation failure within an information basis that is sufficient for the query. Loss is a failure of sufficiency: the projection has discarded task-relevant information needed to distinguish states with different values of q. Equivalently, the projected representation is not a query-relative sufficient statistic for q.

Proposition 3.1 (Projection non-identifiability)**.** *Let $P : S \to R$ be a projection and let $q : S \to A$ be a query. If there exist $s_1, s_2 \in S$ such that*

$$P(s_1) = P(s_2) \qquad \textit{and} \qquad q(s_1) \neq q(s_2),$$

then there is no function $f : R \to A$ satisfying

$$f(P(s)) = q(s)$$

for every $s \in S$.

Proof. Suppose $P(s_1) = P(s_2) = r$ while $q(s_1) \neq q(s_2)$. Any function $f : R \to A$ must assign one value to r. It therefore cannot satisfy both

$$f(r) = q(s_1) \qquad \text{and} \qquad f(r) = q(s_2).$$

Hence no function over the projected representation alone recovers q for every underlying state. □

Proposition 3.1 is elementary. Its role is to mark the boundary between recoverable inference error and representational loss. A determinate answer beyond that boundary requires an additional assumption, representation expansion, or information outside the original operative representation.

Task-relative exact support. The identifiability condition also induces a canonical task-relative quotient of the underlying state space. This construction is closely related to Bruhacs's task-sufficiency and minimal-support treatment of reasoning under projection [4]. For the single query $q : S \to A$, define

$$s_1 \sim_q s_2 \quad \text{iff} \quad q(s_1) = q(s_2),$$

and let

$$Q_q = S/{\sim_q}.$$

The quotient Q_q retains exactly the distinctions required to determine q: states may be collapsed whenever they require the same answer, but not when they require different answers. Any representation sufficient to determine q must therefore distinguish at least the equivalence classes relevant to Q_q, although it may encode those distinctions in a different form.

This provides a useful exact-sufficiency baseline for reasoning under projection. If the operative representation R preserves enough structure to distinguish the relevant classes of Q_q, then the query is identifiable in principle. The primary concern of the present framework is the harder regime in which the operative representation falls below that task-relative threshold, or in which the reasoner cannot establish that the threshold has been met. The question then changes from "what is the exact answer?" to "what response form remains licensed by the support that actually survives?"

Representation-or-form requirement. When a query is not identifiable under the current representation, an admissible response must either change the information basis or change the answer form. It may expand the representation, obtain new information, introduce an explicit assumption or decision rule, weaken the claim, branch over alternatives, mark the dependency as traceable, route the query, or refuse the requested form. What it must not do is present the unsupported determinate answer as if it followed from the original representation alone.

The non-identifiability claim is deliberately limited: no function over the current representation can recover the query for all underlying states. The subsequent formal results concern a different question. They characterize the operational quotient induced by recovery, attribution, and licensing rather than adding further impossibility claims about projection itself.

This separates the descriptive and normative parts of the argument. The descriptive claim is that non-identifiability prevents recovery of q from R alone. The normative claim is an epistemic admissibility criterion: a reasoning system should not present a conclusion as determined by R when the conclusion depends on distinctions that R does not preserve or make reconstructible. This criterion does not require inaction. It requires that action under non-identifiability be marked as a decision rule, assumption, conditional response, request for representation expansion, or unresolved dependency rather than as an inference licensed by the current representation. Non-identifiability does not imply silence; it changes the epistemic status of speech. For brevity, I refer to these moves—representation expansion, external information, explicit assumptions, conditionalization, routing, or refusal of the requested form—as changing the information basis or answer form.

Admissibility is intentionally weaker than correctness. The framework does not determine which decision procedure a system must use when R is insufficient, nor does it prohibit probabilities, confidence intervals, ranked hypotheses, priors, utility functions, or policy choices. It constrains the epistemic status of the output: a claim should not be presented as following from R alone when it depends on distinctions that R does not preserve or make reconstructible. A probabilistic or decision-theoretic response may be admissible if it marks the assumptions, priors, utilities, or decision rules on which it depends; it becomes inadmissible when those elements are presented as if they were determined by the representation itself. This is especially important for AI safety and security because many unsafe outputs are not unsupported in the sense of being nonsensical; they are unsupported in the narrower sense that they present assumptions, priors, tool traces, or policy choices as if they were determined by the available evidence.

Correctness and structural support are distinct axes. Factual correctness and structural reliability should therefore not be treated as equivalent evaluation criteria. A system may use a fully supported epistemic operation and nevertheless make an ordinary inference error. Conversely, a system may issue a determinate claim that is not licensed by the operative representation and happen to guess the correct answer.

Table 1: Factual correctness and structural support are logically distinct.

	Factually correct	**Factually incorrect**
Structurally supported	Justified success	Inference error
Structurally unsupported	Lucky unsupported answer	Unsupported error

The lower-left case is particularly important for evaluation. Suppose $P(s_1) = P(s_2)$ while $q(s_1) \neq q(s_2)$, and a reasoner receiving the common representation $P(s_1) = P(s_2) = r$ nevertheless emits $q(s_1)$. If the underlying state happens to be s_1, ordinary answer accuracy records a success. Yet the same epistemic operation was unsupported in both possible underlying states because the operative representation did not distinguish them. The factual correctness of the realized answer therefore does not retroactively supply the missing representational support.

Thus,

$$\boxed{\text{correct answer} \not\Rightarrow \text{structurally supported answer,}}$$

and

$$\boxed{\text{structurally supported reasoning} \not\Rightarrow \text{correct answer.}}$$

Structural evaluation is concerned with this orthogonal dimension: whether the operation producing the answer was licensed by the support available to the reasoner, not merely whether the realized proposition happened to be true.

Confidence is a further distinct variable. It describes a reasoner's degree of belief or uncertainty about an answer; it does not establish whether the operative representation preserves the distinction required to determine that answer. An identifiable query may receive low confidence because the reasoner is unreliable or uncertain, while a non-identifiable query may receive high confidence because the reasoner fails to recognize the projection boundary. Therefore,

$$\boxed{\text{dependency status} \neq \text{confidence.}}$$

Confidence may help select among already admissible response forms or express uncertainty within one of them, but it cannot enlarge the admissible set or convert a non-identifiable dependency into preserved or reconstructible support.

Immediate consequences. This observation yields three immediate consequences for reasoning under projection.

Unsupported determinacy. If q is not identifiable under P, then any determinate answer to q from R alone must depend on assumptions not contained in R.

Correction boundary. If a failure concerns an identifiable query, correction can in principle operate within the current representation. If the query is not identifiable, correction requires changing the information basis or answer form; otherwise it is not correction but hidden reconstruction.

Decision/inference separation. A system may still act when q is non-identifiable, but the action must be marked as a decision rule under non-identifiability rather than as an inference determined by R.

Operational corollaries. The observation also yields several testable consequences for reasoning systems.

Representation-enrichment monotonicity. If a representation R' refines R by distinguishing states that R collapses, then some queries non-identifiable under R may become identifiable under R'. A projection-aware system should therefore become more willing to make determinate claims when the enriched representation restores the distinction required by the query, while avoiding such claims under the compressed representation.

Traceability is not reconstruction. Adding a provenance pointer, citation identifier, tool-call record, source hash, or packaged research-object relation to R may make a dependency traceable, but it does not make the query identifiable unless the added representation also preserves or recovers the query-relevant distinction. Research-object systems such as RO-Crate demonstrate that artifacts, metadata, provenance, and relationships can be preserved together in a portable representation [26]; the present distinction concerns a separate question: whether the resulting representation contains enough query-relevant structure to recover the dependency on which the claim relies. A system should therefore treat preserved attribution differently from reconstructible support.

Answer-form sensitivity. When q is non-identifiable under R, a direct determinate answer and a conditional answer do not have the same admissibility status. The direct answer presents q as determined by R; the conditional answer marks the additional assumption or decision rule on which the claim depends. The framework therefore predicts that evaluation should score not only content, but whether the answer form matches dependency status.

Diagnostic contrast. A benchmark can test the distinction between error and loss by comparing paired tasks that differ only in whether the query-relevant distinction is represented. If a model gives the same determinate answer form in both the compressed and enriched condition, then it is failing to regulate claims by dependency status. If it withholds, branches, or marks assumptions in the compressed condition and becomes determinate only in the enriched condition, then it is behaving in a projection-sensitive way.

3.6 When Projection Produces Representational Loss

Projection produces representational loss for a query when it collapses a distinction required by that query and the distinction is not otherwise encoded, boundedly reconstructible, or explicitly supplied by a marked assumption. The same projection may remain admissible for other queries whose answers are constant across the collapsed states. Thus loss is query-relative: the representation fails not because it is compressed, but because a downstream claim depends on a distinction the compression does not preserve.

By the representation-or-form requirement, the system must either change the information basis or change the answer form rather than present the claim as determined by the original representation.

4 Classifying Failure: Error vs Representational Loss

The failures described in the previous sections arise from a single structural confusion: the treatment of representational loss as if it were recoverable error. Distinguishing between these two cases is necessary for any reasoning process that operates over projected representations.

Using the notation above, error concerns failures in answering an identifiable query; loss concerns cases where the query is not identifiable from the current representation. The distinction matters because treating the second case as the first leads systems to repair, infer, or explain beyond what the representation supports. The term "loss" is not used here to mean training loss, prediction loss, or a decision-theoretic loss function. It refers to failure of bounded reconstructibility for a query-relevant distinction required by a downstream claim, action, continuation, or decision. Operational invalidity is a separate condition: a representation may support a proposed continuation informationally even though the declared regime does not permit it, and a permitted continuation may still lack adequate informational support.

4.1 Error: Recoverable Deviation Within a Representation

An *error* occurs when a system produces an incorrect result even though the query-relevant distinction remains identifiable and the represented information is sufficient, in principle, to correct the result without changing the information basis. In this case, the failure is due to the reasoning process rather than the representation itself.

Errors are, in principle, correctable. Given the same representation, a different or improved reasoning procedure can recover the correct result. Correction does not require access to information outside the representation; it requires only a revision of how the existing information is used.

Typical examples include:

- arithmetic or logical mistakes,
- incorrect application of rules,
- inconsistent intermediate steps.

In each case, the distinction between correct and incorrect outcomes exists within the representation, and a valid correction can be constructed without introducing new information.

4.2 Loss: Non-Identifiability Under the Current Representation

Representational loss, hereafter loss when context is clear, occurs when the representation no longer contains, or cannot validly reconstruct, the information required to distinguish between multiple operationally relevant underlying states. In this case, no reasoning process operating solely on the current representation can recover the missing distinction without additional assumptions, representation expansion, or external information.

The term "loss" is used here as a query-relative category rather than as a claim that all missing structure has the same cause. A query may become non-identifiable because relevant variables are absent, because multiple causal models remain observationally equivalent, because a metric collapses distinct evaluative dimensions, because provenance is unavailable, or because source-level support is not represented. These mechanisms differ, but they share the same operational consequence: the current representation does not preserve enough structure to determine the requested query without additional assumptions, representation expansion, or external information.

Loss names a structural condition of projection rather than an inference failure: once distinct states have been mapped to the same representation, they are indistinguishable from the perspective of the reasoning system. In this sense, loss is closer to non-identifiability or failure of query-relative sufficiency than to high variance, low confidence, or ordinary prediction error.

Typical examples include:

- multiple causal models that produce identical observable outcomes [21],
- decisions that compress competing values into a single metric [22],
- incomplete information where relevant variables are not represented.

In these cases, there is no uniquely evidentially determined answer that can be derived from the representation alone. A decision rule, prior, utility function, or policy may still select an action, but that selection introduces a basis not supplied by the represented evidence alone. Any attempt to present a single determinate conclusion as evidentially implied by the representation therefore requires introducing assumptions not justified by that information basis.

4.3 Representation-Induced Decision Loss

The categorical distinction between identifiable and non-identifiable queries can be supplemented, when a task admits an explicit loss function, by a quantitative measure of the cost induced by representation. Swanson develops this idea through a best-achievable loss functional over bounded carriers and an excess-risk remainder relative to fuller information [27]. Adapting that construction to the present notation, let $\ell(\hat{a}, q(s))$ measure the task loss associated with answer or action $\hat{a}$ when the relevant target is $q(s)$. Define the best achievable loss using only the projected representation as

$$L(P, q) = \inf_{\phi} \mathbb{E}\left[\ell\big(\phi(P(S)), q(S)\big)\right],$$

where the infimum ranges over admissible decision or reconstruction rules ϕ operating on the representation produced by P.

Relative to reasoning with the unprojected state, define representation-induced excess loss as

$$\Delta L(P, q) = L(P, q) - L(\mathrm{id}, q).$$

This quantity should not be confused with the categorical use of *representational loss* elsewhere in this paper. The categorical notion asks whether the distinctions required by the query survive projection in a form that licenses the requested answer. The quantitative notion asks how much unavoidable task loss is induced by restricting the reasoner to the projected representation. The two are related but not identical.

In particular, non-identifiability need not imply catastrophic task loss: two states collapsed by P may require different exact answers while nevertheless permitting the same low-risk action under a particular loss function. Conversely, a small representational omission may induce large decision loss when the omitted distinction governs a high-consequence branch. The quantitative formulation therefore provides a way to measure the consequence of projection without replacing the dependency-status account of what the representation licenses.

4.4 Representation-Scope and Target-Scope Overreach

The distinction between target formation and representational projection also separates two forms of unsupported reasoning, corresponding closely to Swanson's distinction between rendering overextension and boundary overextension [28].

Representation-scope overreach occurs when the selected target state S_Γ is adequate for the query, but the operative representation R does not preserve or boundedly reconstruct a distinction needed to answer it. The reasoner claims more than the current representation supports.

Target-scope overreach occurs when the declared target state S_Γ itself excludes a distinction material to the query. In this case, even a lossless representation of S_Γ would remain insufficient because the reasoning problem was bounded too narrowly upstream.

The two failures require different repairs. Representation-scope overreach can in principle be addressed by refining P_Γ, retrieving additional information already within the target, or changing answer form. Target-scope overreach requires reconsidering B_Γ: expanding, revising, or contesting what was treated as the relevant system in the first place. A system that recognizes only representational insufficiency may therefore repeatedly enrich the wrong target.

4.5 Misclassification: Treating Loss as Error

Many reasoning failures occur when a system treats loss as if it were error. That is, the system behaves as though additional reasoning steps can recover distinctions that are no longer present in the representation.

This misclassification leads to several characteristic behaviors:

- **Overconfidence:** The system presents a single conclusion without acknowledging indistinguishability, such as naming a root cause when telemetry cannot distinguish among plausible causes.
- **Hallucination:** Missing information is implicitly or explicitly fabricated, such as citing source support that is not present in the retrieved context.
- **Arbitrary resolution:** One of several equivalent possibilities is selected without justification, such as choosing a security fix before the vulnerable code path is represented.
- **Unsafe action:** An agent acts as if authorization, environmental state, user intent, or reversibility is known when those distinctions are not preserved in its current state.

In each case, the system produces an output that appears well-formed, but the reasoning process is no longer grounded in recoverable state.

4.6 Detection and Representation of Loss

Unlike error, loss cannot be corrected through additional reasoning over the same representation alone. The system must instead change the epistemic status of the claim or change the representation on which the claim depends. It may request additional information, expand or refine the representation, reconstruct the distinction through an explicit bounded path, weaken the claim into a conditional, or mark and trace the loss. Marking and tracing preserve accountability for the missing distinction, but they do not make claims depending on that distinction admissible.

Tracing does not recover the missing distinction. It records the dependency, assumption, or point of loss so that downstream claims are not treated as if the distinction were preserved. The response may therefore take the form of ambiguity marking, branching over multiple consistent explanations, separating inference from decision, or routing the query to a source or procedure that can restore the missing distinction.

In some cases, representation expansion is not simply a matter of adding more facts to the current context. The missing distinction may be available only through an external source, measurement procedure, institutional role, policy, or decision-maker. In such cases, an admissible response may route the query to the appropriate source, procedure, role, or authority rather than answer from the current representation. The relevant point is not social authority as such, but dependency status: the current representation does not contain the distinction, and the system must identify where that distinction could be preserved, reconstructed, or validated.

The key requirement is that the system does not collapse indistinguishable states into a single unjustified conclusion.

This makes diagnosability, rather than error elimination, a central reliability objective. A structurally reliable system need not prevent every failure or represent every distinction that could conceivably become relevant. It should preserve enough information about dependencies, ambiguity, provenance, assumptions, and points of loss that materially important failures can, where possible, be localized and distinguished from ordinary inference error. A system that sometimes fails while exposing where its information basis became insufficient may therefore be structurally more reliable than one that produces more correct outputs while leaving its residual failures opaque.

5 Dependency Status as a Thin Waist

For a query q and representation R, *dependency status* refers to the kind of support that R provides for the distinctions on which an answer to q depends. It is not identical to provenance, confidence, observability, or identifiability, although it may involve each of them. Identifiability asks whether a function from R to the answer exists. Dependency status asks what kind of support the current reasoning process has for using that answer: whether the needed distinction is directly preserved, recoverable through a bounded procedure, merely recorded as a dependency, or unavailable even as an auditable relation.

The four statuses arise from a gated recovery–attribution test and are justified behaviorally by the epistemic operations they distinguish after projection. For a claim depending on a distinction d, the system must determine whether d is directly available for inference, recoverable by a bounded procedure, only recorded as a dependency, or unavailable even for audit. These cases license different answer forms. Directly available distinctions can support direct answers. Boundedly recoverable distinctions can support reconstruction or procedural answers. Merely recorded dependencies can support trace marking, routing, or qualification, but not recovery of the distinction itself. Dependencies that are neither recoverable nor traceable cannot support inference or audit from the current representation.

Let $D_q^{(t)}$ denote the dependencies recognized as material to query q at reasoning stage t. This set is provisional: later inquiry may add, split, relate, narrow, or reject dependencies. For notational simplicity, the remainder of this paper writes D_q where the developmental index is not material to the immediate argument.

There is a still earlier assessment problem. Let $D^*_{\Gamma,q}$ denote, model-relatively, the dependencies that are actually material to query q under regime Γ, and let $\widehat{D}_{\Gamma,q}$ denote the dependency structure currently represented or assessed as material by the bounded reasoner. The framework does not assume privileged access to $D^*_{\Gamma,q}$. Projection may therefore produce error both in dependency membership,

$$\widehat{D}_{\Gamma,q} \neq D^*_{\Gamma,q},$$

and in the status assigned to dependencies that have been represented. The operation-licensing machinery governs the represented assessment; it does not certify that every material dependency was discovered or included. A more complete system therefore also requires mechanisms for discovering new dependencies and preserving them while they remain materially operative.

A claim may depend on more than one distinction, and those dependencies need not share the same status. For a query q with relevant dependency structure D_q, dependency status may therefore be represented as a profile

$$\Delta_{\Gamma,q,R} : D_q \to \{\text{preserved}, \text{reconstructible}, \text{traceable}, \text{opaque}\},$$

where $\Delta_{\Gamma,q,R}(d)$ records the status of dependency d relative to the operative representation R and regime Γ. A claim-level description such as "mixed" indicates that this profile is nonuniform; it is not necessarily a fifth primitive rung of the ladder. Response-form selection must preserve the weaker or unresolved dependencies relevant to the particular claim rather than compress the entire profile into an unjustifiably strong scalar status.

A thin-waist interpretation. Dependency status is intended to serve as a compact interface between heterogeneous representations and heterogeneous epistemic operations. Above the interface may sit prompts, retrieved documents, telemetry, memories, metrics, causal models, policy artifacts, intermediate reasoning states, or other domain-specific representations. Below it may sit direct inference, bounded reconstruction, retrieval, qualification, routing, reframing, explicit decision, deferral, or refusal of the requested form. These systems need not share one substantive ontology. They meet at the narrower question of what kind of support the operative representation retains for the dependency required by the proposed continuation.

Schematically,

heterogeneous representations and dependency structures
$\downarrow$
dependency-status interface
$\downarrow$
heterogeneous epistemic operations and response forms.

The interface is intentionally smaller than a complete theory of reasoning. It does not encode the full meaning of the represented objects, choose an optimal continuation, or certify that every material dependency has been discovered. Its purpose is to preserve the distinctions required to prevent the downstream operation from acquiring a stronger epistemic license than the represented dependency structure supports.

Calling dependency status a thin waist does not make the four statuses a universal ontology. A particular domain may refine them with probabilistic, partial, contested, provenance-conflicted, authority-relative, or other qualifiers. A particular controller may also behave identically toward statuses that its available operation repertoire cannot distinguish. The relevant theoretical question is therefore not whether four labels exhaust every possible epistemic condition, but which distinctions an interface must retain to preserve operation licensing across the contexts for which it is intended.

Structural type and dependency status. Dependency status describes what happened to a query-relevant dependency under the operative representation, but it does not by itself describe what kind of structure the dependency carries. Following the task-structure distinction developed by Bruhacs [4], it is useful to distinguish at least three broad structural roles:

$$\tau_q(d) \in \{\text{invariant}, \text{selector}, \text{operator}\}.$$

An *invariant dependency* concerns a property or distinction whose value must survive for the query to remain supported. A *selector dependency* concerns structure needed to choose among alternatives, interpretations, weights, models, or continuations. An *operator dependency* concerns the conditions needed to license a transformation, inference rule, procedure, or reasoning operation.

Structural type and dependency status are orthogonal. A dependency may therefore be represented by a pair

$$\big(\tau_q(d), \Delta_{\Gamma,q,R}(d)\big).$$

For example, an invariant may be preserved while an operator condition is merely traceable; or a selector may be reconstructible while the provenance needed to justify that reconstruction remains opaque. This prevents the dependency ladder from collapsing distinct kinds of structural loss into a single scalar notion of support.

Bruhacs further shows that inferential or explanatory machinery can require hidden support even where term-level or state-level analysis is insufficient [2]. For the present framework, this motivates making the operator distinction explicit. Let

$$D^{S}_{\Gamma,q}$$

denote dependencies concerning represented state or content, and let

$$D^{\Omega}_{\Gamma,q}$$

denote dependencies concerning the applicability of an operation, inference rule, procedure, or transformation. These sets may overlap in their underlying evidence but play different licensing roles.

A system may therefore possess sufficient represented content to execute an operation while lacking sufficient represented structure to establish that the operation is applicable. Conversely, the conditions licensing an operation may be represented even when the object-level state required to complete it is missing. Thus

$$D^{S}_{\Gamma,q} \text{ sufficient } \not\Rightarrow D^{\Omega}_{\Gamma,q} \text{ sufficient.}$$

Successful execution is consequently not evidence, by itself, that the application was licensed. This distinction becomes important for reusable reasoning scaffolds, inherited procedures, diagnostic rules, and agentic policies whose operational form may survive projection more readily than the contextual conditions governing their use.

Operational invalidity is not another rung on this ladder. Dependency status records the kind of informational support available for a claim. Operational validity separately concerns whether the represented state or proposed continuation is permitted within the declared regime. A system must preserve both judgments rather than compressing operational failure into an informational dependency category.

Thus the term *ladder* refers to an operational weakening of available response forms, not to a claim that all support relations form a total epistemic order or one scalar dimension. The formal results below distinguish three questions: whether primitive support conditions factor through the four base statuses, whether those statuses may be compressed across a family of licensing contexts, and whether one particular controller must maintain all four distinctions internally. More fine-grained conditions such as probabilistic reconstruction, partial reconstruction, conflicting provenance, or prior-dependent inference may refine the base interface whenever the declared operation family distinguishes them.

Definition 5.1 (Primitive support condition and recovery mode)**.** Fix an assessment context

$$\eta = (\Gamma, q, R),$$

consisting of an operating regime, query, and operative representation. Let X_η be a space of primitive support conditions for dependencies treated as material in that context. An element

$x \in X_\eta$ is not itself a dependency. It records the context-relative condition of support for a dependency.

Define the recovery-mode map

$$\rho_\eta : X_\eta \to \{\mathsf{direct}, \mathsf{bounded}, \mathsf{unavailable}\}.$$

The value direct means that the query-relevant distinction is directly available in R. The value bounded means that the distinction can be restored by a procedure permitted within the resource, interaction, and authority bounds of Γ. The value unavailable means that no such direct or bounded recovery is available in the declared assessment context.

Attribution becomes status-resolving only when recovery is unavailable. Accordingly, define the partial attribution map

$$\alpha_\eta : \rho_\eta^{-1}\big(\{\mathsf{unavailable}\}\big) \to \{\mathsf{attributable}, \mathsf{unattributable}\}.$$

The attribution criterion is supplied independently as part of the declared regime and without reference to κ_η or λ_η. A condition is attributable when the operative representation preserves a typed identity or relation that localizes the missing dependency's source, object, or governing role sufficiently to distinguish that dependency from an unspecified absence and to preserve an accountable relation for possible follow-up. This structure may enable inspection, retrieval, routing, contestation, or audit when the corresponding capability is available. Attribution does not require the current controller to possess such a capability, and it does not imply that the query-relevant distinction or source content can be recovered.

Definition 5.2 (Gated dependency-status map)**.** Let

$$\Sigma = \{\mathsf{P}, \mathsf{R}, \mathsf{T}, \mathsf{O}\}$$

denote the base dependency-status space. For a fixed assessment context η, define

$$\kappa_\eta : X_\eta \to \Sigma$$

by

$$\kappa_\eta(x) = \begin{cases} \mathsf{P}, & \rho_\eta(x) = \mathsf{direct}, \\ \mathsf{R}, & \rho_\eta(x) = \mathsf{bounded}, \\ \mathsf{T}, & \rho_\eta(x) = \mathsf{unavailable} \text{ and } \alpha_\eta(x) = \mathsf{attributable}, \\ \mathsf{O}, & \rho_\eta(x) = \mathsf{unavailable} \text{ and } \alpha_\eta(x) = \mathsf{unattributable}. \end{cases}$$

The symbols correspond respectively to preserved, reconstructible, traceable, and opaque dependency status.

Definition 5.2 yields the conditional recovery–attribution structure

$$\Sigma \cong \{\mathsf{direct}, \mathsf{bounded}\} \sqcup \left(\{\mathsf{unavailable}\} \times \{\mathsf{attributable}, \mathsf{unattributable}\}\right).$$

This is a gated construction rather than an unrestricted product. Attribution distinguishes the residual conditions specifically after recovery has failed. Traceable status is therefore not an intermediate quantity of reconstruction. It records a different residual capability: the distinction cannot presently be recovered, but the missing dependency can still be localized, inspected, retrieved, routed, contested, or governed in ways unavailable under opaque status.

Definition 5.3 (Nested capability thresholds)**.** For the primitive support space X_η, define

$$E_{1,\eta} = \{x \in X_\eta : \rho_\eta(x) = \mathsf{direct}\},$$

$$E_{2,\eta} = \{x \in X_\eta : \rho_\eta(x) \in \{\mathsf{direct}, \mathsf{bounded}\}\},$$

and

$$E_{3,\eta} = E_{2,\eta} \cup \{x \in X_\eta : \rho_\eta(x) = \mathsf{unavailable} \text{ and } \alpha_\eta(x) = \mathsf{attributable}\}.$$

These sets satisfy $E_{1,\eta} \subseteq E_{2,\eta} \subseteq E_{3,\eta} \subseteq X_\eta$ by construction. Their four regions are

$$E_{1,\eta}, \qquad E_{2,\eta} \setminus E_{1,\eta}, \qquad E_{3,\eta} \setminus E_{2,\eta}, \qquad X_\eta \setminus E_{3,\eta},$$

corresponding respectively to $\mathsf{P}, \mathsf{R}, \mathsf{T}, \mathsf{O}$. Define the associated capability signature by

$$\chi_\eta(x) = \big(\mathbf{1}_{E_{1,\eta}}(x), \mathbf{1}_{E_{2,\eta}}(x), \mathbf{1}_{E_{3,\eta}}(x)\big).$$

Its possible values are

$$(1,1,1), \qquad (0,1,1), \qquad (0,0,1), \qquad (0,0,0),$$

and κ_η is obtained by naming these signatures $\mathsf{P}, \mathsf{R}, \mathsf{T}, \mathsf{O}$, respectively.

The nested construction does not introduce primitives independent of ρ_η and α_η. It reorganizes the same gated recovery–attribution conditions as three separately testable capability thresholds. The inclusions are automatic; surjectivity of κ_η is what ensures that all four regions are nonempty and hence that all four signatures are realized.

Remark 5.4 (Base statuses and refinements)*.* The gated map defines a base response-governance interface. It does not claim that attribution, probability, provenance quality, authority, or contestation become irrelevant whenever recovery succeeds. If a declared operation family distinguishes, for example, directly available support with adequate provenance from directly available support without adequate provenance, the interface must retain that distinction as a qualifier or refinement. The four-class claim below therefore concerns the declared base operation family for which licensing depends on the gated recovery–attribution condition.

Let O_{base} denote that declared base family of epistemic operations, and let L contain the licensing outcomes relevant to the regime. Before introducing any status-based coarsening, define primitive licensing by

$$\lambda_\eta : X_\eta \times O_{\text{base}} \to L.$$

The primitive licensing relation is specified through the requirements of the operations themselves. Direct inference may require direct availability; reconstruction requires a permitted bounded recovery path; provenance-following or routing requires an adequate attributable relation; and qualification, deferral, or refusal of the requested form need not require recovery. Operational validity of an associated action remains a separate condition developed below.

Call λ_η *status-sufficient* when

$$\kappa_\eta(x) = \kappa_\eta(y) \quad \Longrightarrow \quad \forall o \in O_{\text{base}},\ \lambda_\eta(x,o) = \lambda_\eta(y,o).$$

Status sufficiency says that the gated map does not discard a primitive distinction that the declared base licensing relation still uses.

Because χ_η and κ_η have exactly the same fibers, status sufficiency may equivalently be stated as capability sufficiency:

$$\chi_\eta(x) = \chi_\eta(y) \quad \Longrightarrow \quad \forall o \in O_{\text{base}},\ \lambda_\eta(x,o) = \lambda_\eta(y,o).$$

This is an equivalent presentation of the sufficiency hypothesis, not an independently weaker assumption.

Proposition 5.5 (Licensing factorization)**.** *Suppose that κ_η is surjective and that λ_η is status-sufficient. Then there exists a unique status-level licensing relation*

$$\Lambda_\eta : \Sigma \times O_{\text{base}} \to L$$

such that

$$\lambda_\eta(x,o) = \Lambda_\eta\big(\kappa_\eta(x), o\big)$$

for every $x \in X_\eta$ and $o \in O_{\text{base}}$.

Proof. For $s \in \Sigma$, choose any $x \in X_\eta$ satisfying $\kappa_\eta(x) = s$, and define

$$\Lambda_\eta(s,o) = \lambda_\eta(x,o).$$

Surjectivity guarantees that such an x exists. Status sufficiency guarantees that the value is independent of which member of the fiber $\kappa_\eta^{-1}(s)$ is chosen, so the definition is well-defined. The factorization follows immediately. Surjectivity also makes the induced relation unique. □

The factorization proposition says that dependency status is sufficient for the declared licensing behavior. It does not yet show that all four status distinctions are necessary. For that, the licensing relation must also distinguish different fibers of κ_η.

Define primitive behavioral equivalence by

$$x \approx_\eta y \quad \Longleftrightarrow \quad \forall o \in O_{\text{base}},\ \lambda_\eta(x,o) = \lambda_\eta(y,o).$$

Call the primitive licensing relation *fiber-separating* when

$$\kappa_\eta(x) \neq \kappa_\eta(y) \quad \Longrightarrow \quad \exists o \in O_{\text{base}} \quad \lambda_\eta(x,o) \neq \lambda_\eta(y,o).$$

Definition 5.6 (Capability-threshold witness)**.** For $k \in \{1,2,3\}$, an operation $o_k \in O_{\text{base}}$ is a witness for $E_{k,\eta}$ when

$$x \in E_{k,\eta},\ y \notin E_{k,\eta} \quad \Longrightarrow \quad \lambda_\eta(x,o_k) \neq \lambda_\eta(y,o_k).$$

A threshold-witnessing family consists of one such operation for each of the three capability thresholds.

The witnesses concern capabilities rather than status-preferred responses. For example, the relevant thresholds may concern direct determination, determination through a permitted bounded procedure, and accountable handling of a localized dependency. They need not coincide with whichever operation a policy would prescribe after the status has been assessed.

Proposition 5.7 (Threshold witnesses imply fiber separation)**.** *If $E_{1,\eta}$, $E_{2,\eta}$, and $E_{3,\eta}$ admit a threshold-witnessing family, then λ_η is fiber-separating.*

Proof. If $\kappa_\eta(x) \neq \kappa_\eta(y)$, then the corresponding nested capability signatures satisfy $\chi_\eta(x) \neq \chi_\eta(y)$. Hence their membership differs for at least one threshold $E_{k,\eta}$. The witness o_k for that threshold satisfies $\lambda_\eta(x, o_k) \neq \lambda_\eta(y, o_k)$, so the two primitive support conditions are behaviorally distinguishable. □

Theorem 5.8 (Primitive behavioral completeness)**.** *Suppose that:*

1. *$\kappa_\eta : X_\eta \to \Sigma$ is surjective;*
2. *λ_η is status-sufficient; and*
3. *λ_η is fiber-separating.*

Then

$$x \approx_\eta y \quad \iff \quad \kappa_\eta(x) = \kappa_\eta(y),$$

and consequently

$$X_\eta/{\approx_\eta} \cong \Sigma.$$

Proof. If $\kappa_\eta(x) = \kappa_\eta(y)$, status sufficiency gives $x \approx_\eta y$. If $\kappa_\eta(x) \neq \kappa_\eta(y)$, fiber separation supplies an operation whose licensing outcomes differ, so $x \not\approx_\eta y$. Behavioral equivalence therefore has exactly the fibers of κ_η as its equivalence classes. Because κ_η is surjective, the induced map from $X_\eta/{\approx_\eta}$ to Σ is a bijection. □

Corollary 5.9 (Witnessed primitive behavioral completeness)**.** *Suppose that κ_η is surjective, λ_η is status-sufficient, and the three nested capability thresholds admit a threshold-witnessing family. Then*

$$X_\eta/{\approx_\eta} \cong \Sigma.$$

Proof. Proposition 5.7 supplies fiber separation, after which Theorem 5.8 applies. □

Theorem 5.8 gives the general primitive-to-status result, while Corollary 5.9 supplies a concrete sufficient route through three threshold-witnessing operations. The nested construction makes the separation hypothesis inspectable: different status regions cross different recovery or attribution thresholds, and a declared operation family preserves all four behavioral classes when it contains operations that witness those crossings. This is a reorganization and partial strengthening of the theorem's hypotheses, not a derivation from primitives independent of the gated recovery–attribution construction. The result remains conditional on the recovery and attribution semantics, operation family, and assessment context; it is not a claim that reality contains exactly four epistemic states.

The number of statuses a particular controller must retain can still be smaller. Using the induced licensing relation from Proposition 5.5, define local status-level equivalence by

$$s \equiv_\eta t \quad \iff \quad \forall o \in O_{\text{base}},\ \Lambda_\eta(s, o) = \Lambda_\eta(t, o).$$

A controller may compile locally equivalent statuses into one internal control state without changing its present licensing behavior. For example, a controller with no retrieval, routing, provenance-following, or escalation operation may treat traceable and opaque dependencies identically.

Local equivalence does not establish that the same compression is safe across controller or regime boundaries. For a declared family of status-level licensing contexts $\mathfrak{C}$, define

$$s \equiv_{\mathfrak{C}} t \quad \Longleftrightarrow \quad \forall \eta \in \mathfrak{C},\ \forall o \in O_{\text{base}},\ \Lambda_\eta(s,o) = \Lambda_\eta(t,o).$$

Equivalently,

$$\equiv_{\mathfrak{C}} = \bigcap_{\eta \in \mathfrak{C}} \equiv_\eta .$$

The quotient

$$\Sigma/{\equiv_{\mathfrak{C}}}$$

is the coarsest status abstraction preserving the declared status-level licensing behavior across $\mathfrak{C}$.

Call $\mathfrak{C}$ *status-separating* when every pair of distinct statuses has a licensing witness:

$$\forall s \neq t \in \Sigma,\ \exists \eta \in \mathfrak{C},\ \exists o \in O_{\text{base}} \quad \Lambda_\eta(s,o) \neq \Lambda_\eta(t,o).$$

Theorem 5.10 (Portable status non-collapse)**.** *If $\mathfrak{C}$ is status-separating, then*

$$\Sigma/{\equiv_{\mathfrak{C}}} \cong \Sigma.$$

Moreover, no proper quotient of Σ preserves every declared licensing relation in $\mathfrak{C}$.

Proof. Status separation implies that no two distinct members of Σ are equivalent under $\equiv_{\mathfrak{C}}$. Consequently, $\equiv_{\mathfrak{C}}$ is equality on Σ, and its quotient is isomorphic to Σ.

Now suppose that a proper quotient identifies distinct statuses s and t. Status separation supplies a context and operation for which their licensing outcomes differ. Any licensing relation factored through that quotient would have to assign the same outcome to the identified statuses, contradicting the separating witness. □

Thus,

$$\boxed{\text{local behavioral equivalence } \not\Rightarrow \text{ portable behavioral equivalence}}$$

and

$$\boxed{\text{safe compression for current control } \not\Rightarrow \text{ safe compression for destination control.}}$$

Remark 5.11 (Three minimality questions)*.* The results distinguish three claims.

1. *Primitive behavioral completeness* asks whether primitive support conditions factor through exactly the four base status classes for the declared operation family. Theorem 5.8 answers this under surjectivity, status-sufficiency, and fiber-separation assumptions, while Corollary 5.9 shows that three capability-threshold witnesses are sufficient to establish the required separation.

2. *Portable status non-collapse* asks whether those statuses may be identified across a family of status-level licensing contexts. Theorem 5.10 shows that no proper quotient preserves a status-separating family.

3. *Implementation minimality* asks whether a particular controller must literally maintain four internal states. It need not do so when its local operation repertoire treats some statuses identically.

The status assigned in one assessment context does not certify the status that obtains in another. A destination with different evidence, recovery tools, authority, resource bounds, or query may need to reassess ρ_η, α_η, and κ_η. Portable status preservation prevents premature interface-level collapse; it does not replace destination-relative assessment.

The portable quotient also matters when assessments cross interfaces or several represented reasoning objects are combined. A distinction that is behaviorally irrelevant to one controller may remain necessary to another controller or future context. The present results therefore justify dependency status as a control interface while preserving the distinction between primitive support assessment, portable interface non-collapse, and controller-local implementation.

In information-theoretic language, the ladder distinguishes different kinds of information support after projection. Preserved distinctions remain directly encoded in R. Reconstructible distinctions are recoverable by a bounded decoding or derivation procedure from R. Traceable distinctions retain attributable structure—such as provenance, dependency relations, source identities, packaged artifacts, or pointers—without preserving or boundedly recovering enough query-relevant structure to reconstruct the distinction required by the claim. Opaque distinctions are absent even as auditable dependencies. The ladder therefore separates having information about a dependency from having enough information to recover the distinction on which the claim depends.

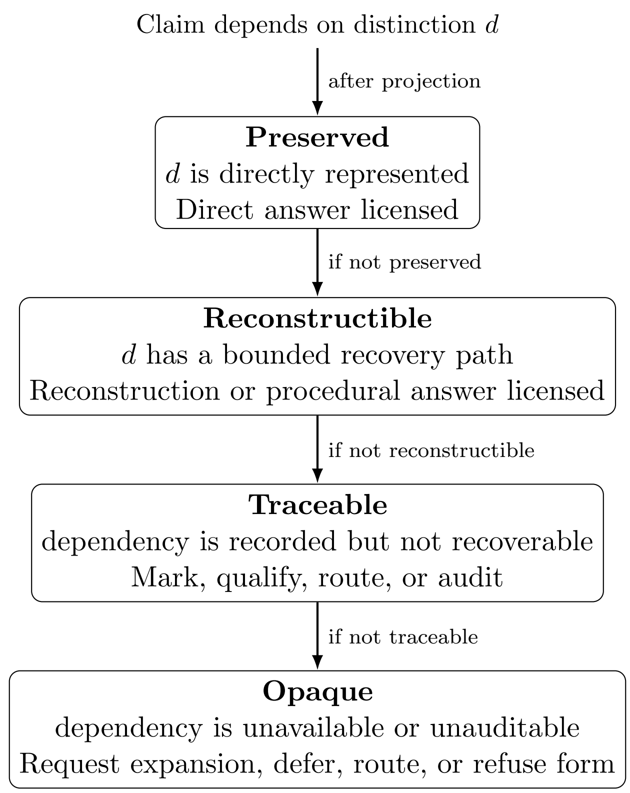

Figure 1: Operational dependency-status pathway under projection. The diagram shows the response-governance cases used by the framework, not a claim that all epistemic support relations form a total order. As recoverability and auditability weaken, the strongest licensed answer form also weakens.

Figure 1 visualizes the operational pathway described in Table 2. It should be read as a decision sequence—directly available, otherwise boundedly recoverable, otherwise attributable without recovery, otherwise opaque—rather than as a claim that every possible support relation lies on one scalar epistemic axis.

Table 2: Illustrative dependency-status licensing matrix.

Status	**Infer**	**Reconstruct**	**Retrieve/trace**	**Qualify/reframe**	**Refuse form**
Preserved	L	C	C	C	C
Reconstructible	U[†]	L	C	L	C
Traceable	U	U	L	L	L
Opaque	U	U	U	L	L

L = licensed, C = conditionally available, and U = unlicensed from the current support state. [†]Direct inference is not licensed while the dependency remains reconstructible. Successful bounded recovery changes the operative information state so that the recovered distinction is preserved, after which direct inference may become licensed. The matrix is regime- and query-relative; it illustrates the operation partition rather than prescribing a universal domain-independent policy. An operation that expands the information basis may move an opaque dependency into a traceable or recoverable state, but that transition does not make targeted retrieval or trace-following licensed under the original opaque state.

The matrix specializes the licensing relation introduced above to the three outcomes used in this illustration:

$$\Lambda_\eta(\delta, o) \in \{\text{licensed}, \text{conditional}, \text{unlicensed}\},$$

where δ is the status of an individual dependency and o is an epistemic operation. For a single dependency status, the fully licensed operation set is therefore

$$\mathcal{O}^{\text{ep}}_\eta(\delta) = \{o \in O_{\text{base}} : \Lambda_\eta(\delta, o) = \text{licensed}\}\,.$$

At the claim level, write

$$\mathcal{O}^{\text{ep}}_\eta(\Delta)$$

for the epistemic operation set licensed by the relevant dependency-status profile rather than by any one dependency considered in isolation. No scalar aggregation rule over the profile is assumed. A claim may have dependencies with different statuses, and an operation is licensed only when the dependencies material to that operation have the required support. The weakest status in the profile therefore need not govern every continuation; it governs those continuations that depend on it. Operational validity supplies a separate gate, formalized below.

The key boundary is between recovery and attribution. A traceable dependency preserves an accountable relation to the missing support without preserving enough information to recover the query-relevant distinction itself.

The boundary between reconstructible and traceable depends on whether the representation contains enough information to recover the distinction needed by the query, not merely enough information to identify where that distinction once came from. For example, a source hash, citation identifier, or provenance pointer may make a dependency traceable by recording that a claim depends on an external source. It is reconstructible only if the representation available to the reasoner also includes, retrieves, or otherwise supplies enough source content and recovery rules to determine the query-relevant distinction. If the pointer records that support exists but the relevant content is unavailable, the dependency is traceable rather than reconstructible. If even the dependency or source relation is not represented, the status is opaque.

This distinction is central for retrieval-augmented and tool-using AI systems. A retrieved document identifier, citation, tool-call trace, or audit log may make a dependency traceable without making the claim reconstructible. If the system cannot access the relevant source content, execution state, or recovery rule, then it should not treat the pointer as support for the claim. In safety and security settings, confusing traceability with reconstructibility can produce false assurance: the system appears grounded because it has a pointer, but the representation available to the reasoner does not actually support the conclusion.

This partition separates three questions that should remain distinct:

$$\boxed{\text{identifiability} \neq \text{recoverability} \neq \text{admissibility.}}$$

Identifiability is an existence condition: if q is identifiable under P, then some function from R to the required answer exists. *Recoverability* is a reasoner- and regime-relative operational condition: the required distinction can actually be restored by an available recovery procedure using only the sources, tools, assumptions, and transformations permitted by the declared regime Γ, within its stated resource or interaction bounds. *Admissibility* is a licensing condition: given the represented support and available recovery capability, the proposed epistemic operation belongs to the operation set permitted for the query and regime.

Accordingly, a distinction is *boundedly reconstructible* only relative to such a declared recovery regime. Existence of an abstract decoding function does not by itself make a dependency reconstructible for a particular system. A query may therefore be identifiable in principle while not being recoverable by the current reasoner, and a recoverable distinction need not license a direct answer until the required reconstruction has actually been performed or otherwise represented as satisfied.

Preserved and reconstructible distinctions can support inference. Traceable status preserves an accountable attribution relation to the missing dependency, but not recovery of the query-relevant distinction required by the claim. Opaque status supports neither such recovery nor accountable localization from the current representation. This ladder is therefore not a complete theory of provenance or explanation; it specifies what kind of support a downstream claim has after projection.

Dependency status, relative to the dependencies currently recognized as material, is the structural relation that later requirements are designed to preserve across reasoning steps, representation changes, correction attempts, and answer-form selection. Observability makes possible the detection or marking of unsupported dependency status. Bounded correction prevents a system from changing dependency status silently during repair. Non-collapse prevents claims supported at one level, such as a representation, intermediate state, or selected explanation, from being treated as if they were supported at another. The requirements that follow should therefore be read not as independent virtues, but as constraints on preserving dependency status under reasoning. When reasoning discovers a new dependency or a new relation among existing objects, the relevant invariant is not preservation of the earlier graph unchanged. It is preservation of the newly discovered constraint and its effect on subsequent licensing.

A later projection that preserves a claim while omitting a newly discovered material constraint may therefore license an answer form stronger than the revised reasoning state supports. This is a form of *silent constraint erasure*: the claim survives while a constraint that limits its legitimate use disappears.

Real systems may attach qualifiers such as probabilistic, partial, prior-dependent, or provenance-conflicted to these base statuses. Such qualifiers refine the operational partition rather than converting it into a total ordering.

6 From Dependency Status to Licensed Epistemic Operation

Because admissibility constrains epistemic operation, dependency status determines not only what claims may be presented as supported but what operations may legitimately precede or replace such a claim. Preserved support may license direct inference; reconstructible support may license a bounded reconstruction followed by inference; traceable support may license retrieval, routing, qualification, or explicit dependency marking; and opaque support may require representation expansion, deferral, or refusal of the requested form. A proposed action or continuation must additionally satisfy the operational validity conditions of the declared regime. The framework therefore governs both the epistemic operation used to obtain or communicate a result and, separately, whether an associated continuation is operationally permitted.

Definition 6.1 (Unsupported promotion)**.** An *unsupported promotion* occurs when a reasoning process treats a weaker, differently typed, or merely retained support relation as though it established the additional conditions required for a stronger epistemic license, without an explicit represented witness for that strengthening.

The three structural requirements act as controls against this common failure mechanism. Observability helps expose candidate promotions, bounded correction prevents hidden strengthening during repair, and non-collapse prevents support from being transferred illegitimately across representational roles or levels.

Schematically,

$$\boxed{\text{weaker or differently typed support } \not\Rightarrow \text{ stronger epistemic license.}}$$

Several recurrent failures are instances of this pattern:

$$\text{traceable} \not\Rightarrow \text{reconstructible,}$$

$$\text{policy-selected} \not\Rightarrow \text{evidence-supported,}$$

and

$$\text{procedure preserved or executable} \not\Rightarrow \text{procedure licensed.}$$

A promotion may be justified, but the additional licensing conditions must themselves be preserved, boundedly reconstructed, retrieved, or introduced through an explicit change of information basis. Mere persistence of a pointer, selection, outcome, procedure, or successful execution is not such a witness.

This principle also explains why omission can strengthen a later representation only illegitimately. If a constraint, qualifier, provenance relation, or operator condition is removed while the dependent claim or procedure is retained, the resulting representation must not inherit a stronger license merely because the limiting structure is no longer visible.

Unsupported promotion therefore provides a common description of several otherwise different failure classes analyzed throughout this paper: treating loss as recoverable error, treating traceability as reconstruction, treating policy selection as evidential determination, treating a proxy or intermediate representation as the underlying target, and treating preservation or successful execution of a procedure as evidence that its application remains licensed. The surface behaviors differ, but the structural error is the same: a downstream continuation acquires a stronger epistemic role than the represented support warrants.

6.1 Licensing, Operational Validity, Admissibility, and Selection

In this section, an *epistemic operation* is a reasoning or communicative move such as inference, reconstruction, qualification, or retrieval. An *operational continuation* is the larger system move that may perform, communicate, or act on an epistemic result. The notation $\mathcal{O}_\eta$ ranges over candidate continuations: $\mathcal{O}_\eta^{\mathrm{ep}}$ retains those whose epistemic component is licensed by dependency status, while $\mathcal{O}_\eta^{\mathrm{valid}}$ separately retains those permitted by the governing regime. A purely epistemic continuation may have a trivial operational-validity condition; an action-bearing continuation need not.

Dependency status constrains epistemic licensing, but epistemic licensing is not the only condition on a continuation. In assessment context $\eta = (\Gamma, q, R)$, the dependency-status profile determines the epistemically licensed subset

$$\mathcal{O}_\eta^{\mathrm{ep}}(\Delta) \subseteq \mathcal{O}_\eta.$$

Operational validity supplies a separate subset:

$$\mathcal{O}_\eta^{\mathrm{valid}} = \{o \in \mathcal{O}_\eta : \mathrm{Valid}_\Gamma(R, o)\}.$$

Definition 6.2 (Admissible continuation)**.** A continuation is admissible in context η under dependency profile Δ when it is both epistemically licensed and operationally valid. Thus

$$\mathcal{O}_\eta^{\mathrm{adm}}(\Delta) = \mathcal{O}_\eta^{\mathrm{ep}}(\Delta) \cap \mathcal{O}_\eta^{\mathrm{valid}}.$$

The framework does not derive operational validity from dependency status. The predicate $\mathrm{Valid}_\Gamma(R, o)$ is an externally parameterized, regime-relative constraint supplied by the governing environment. The representation R appears as an argument because the governing regime may evaluate validity using represented facts about authorization, resources, system state, reversibility, or execution conditions; those facts are inputs to a regime-supplied validity test, not evidence from which operational permission is inferred. The predicate may therefore represent authority, safety, resource, execution, reversibility, or other operational conditions without treating those conditions as consequences of evidential support. The two gates answer different questions. Epistemic licensing asks whether the represented dependency structure supports the proposed kind of inference, reconstruction, qualification, retrieval, or response. Operational validity asks whether the associated continuation is permitted under the declared regime. Neither gate subsumes the other.

A compliant selection policy prescribes only from the admissible set. Fix an assessed dependency-status profile $\widehat{\Delta}_{\Gamma,q,R}$. Let B denote the applicable budget or resource conditions and let Π denote the declared selection policy. Write $o_{\mathrm{prescribed}}$ for the continuation prescribed by that policy:

$$o_{\mathrm{prescribed}} \in \mathrm{Select}\left(\mathcal{O}_\eta^{\mathrm{adm}}\big(\widehat{\Delta}_{\Gamma,q,R}\big), B, \Pi\right).$$

Budget, utility, urgency, reversibility, and policy may distinguish among already admissible continuations, but they do not enlarge the admissible set by manufacturing missing support or operational permission. A low-cost continuation does not become epistemically licensed merely because it is inexpensive, and a high-utility action does not become evidence-determined or operationally valid merely because policy favors it.

The same distinction governs stopping. When a terminal response is admissible, stopping is permitted; it does not follow that stopping is optimal. Further investigation may itself be admissible when its expected diagnostic value justifies its cost. It may resolve a dependency shared by several questions, create reusable recovery infrastructure, reduce later uncertainty, or preserve options that premature closure would remove. Conversely, the existence of some admissible continuation does not imply that a terminal answer is licensed: retrieval, clarification, representation expansion, or routing may be the only currently admissible next operation.

At portfolio scale, several active questions or failures may therefore contribute policy-relevant pressure toward examination of one shared dependency. Such aggregation may affect attention and investigation priority, but it does not strengthen the dependency's epistemic status, establish operational permission or authority, or automatically resolve the affected items if that dependency is later corrected.

Accordingly,

$$\boxed{\text{admissibility} \neq \text{selection} \neq \text{budget allocation.}}$$

Admissibility bounds the available choice set. Selection and budget policies prescribe which admissible continuation should be taken and how long inquiry should continue. Actual system behavior may comply with or depart from that prescription. The framework therefore does not prescribe either immediate abstention or indefinite investigation.

Classification is likewise not compulsion. The framework may classify an operation as admissible or inadmissible without guaranteeing that a reasoner will comply. An implementation may merely report the classification, encourage compliance through prompting, record violations for audit, gate selected transitions, or enforce the constraint mechanically. These are different governance strengths. The operation actually selected must therefore remain distinct from the policy-prescribed operation, the dependency-status assessment, and the admissible set induced by that assessment.

Respecting these constraints does not make reasoning systems infallible. It changes the status of their failures. Unsupported claims become visible rather than hidden; ambiguity is represented rather than suppressed; correction remains tied to the information basis of the representation; and decisions under non-identifiability are marked as decisions rather than inferred conclusions.

For AI reasoning systems, especially safety-critical ones, this changes what counts as a reliable answer. A reliable system should not merely be fluent, decisive, or consistent with a preferred output format. It should calibrate the form of its answer to the dependency status of the claim: direct when support is preserved, procedural when support is reconstructible, conditional when assumptions are required, trace-marked when support is external, and blocked or rerouted when the dependency is opaque.

6.2 Actual, Assessed, Reported, Prescribed, and Selected State

Runtime governance requires a distinction between actual, assessed, and reported dependency status, together with the operation prescribed by a compliant policy and the operation actually selected by the system. For each dependency d under evaluation, let $x^*_\eta(d) \in X_\eta$ denote the primitive support condition that semantically obtains in assessment context $\eta = (\Gamma, q, R)$ under the declared recovery and attribution semantics. Define the actual, or reference, dependency status by $\delta^*_{\Gamma,q,R}(d) = \kappa_\eta\big(x^*_\eta(d)\big)$. This is a semantic property of the relation among the dependency, operative representation, recovery conditions, query, and regime; it need not be observable, computable, or correctly assessable by the bounded reasoner operating on R. Let $\widehat{\delta}_{\Gamma,q,R}(d)$ denote the status assessed by the reasoner, and let $\widetilde{\delta}_{\Gamma,q,R}(d)$ denote the status reported in the resulting answer or reasoning artifact. Collectively, these induce actual, assessed, and reported dependency-status profiles $\Delta^*_{\Gamma,q,R}$, $\widehat{\Delta}_{\Gamma,q,R}$, and $\widetilde{\Delta}_{\Gamma,q,R}$ over the dependencies currently treated as material to the query. The normative framework constrains response form relative to the reference profile, while practical systems can act only through imperfect assessed profiles. Evaluation must therefore consider both whether the assessed profile is accurate and whether the reported answer form faithfully reflects that assessment.

These three quantities should not be identified. In particular,

$$\widehat{\delta}_{\Gamma,q,R}(d) \neq \delta^{*}_{\Gamma,q,R}(d)$$

is possible even when the reasoner follows its assessment procedure correctly. The operative representation may itself omit the evidence needed to determine whether a dependency is preserved, reconstructible, traceable, or opaque. A bounded reasoner can therefore be locally correct relative to its represented evidence while being mistaken about the adequacy of that evidence. Neighboring impossibility results reinforce the general concern: Bruhacs shows that the observational or public representation alone does not in general determine the appropriate hidden reconstruction across heterogeneous tasks [3]. The distinction introduced here is different: it asks how that underdetermination affects a reasoner's assessment and report of its own dependency status.

This creates a second-order licensing problem. First-order epistemic licensing asks whether the answer form is licensed by the actual dependency status. Second-order licensing asks what the reasoner is licensed to claim about its own assessment of that status. Successful execution of a governance procedure does not certify that the representation supplied enough structure for the procedure to classify its own applicability correctly.

The reported status introduces a further requirement:

$$\widetilde{\delta}_{\Gamma,q,R}(d) \preceq_{\eta} \widehat{\delta}_{\Gamma,q,R}(d),$$

Here $a \preceq_{\eta} b$ means that every represented-support commitment communicated by report a is licensed by assessed status b in context η. This is a context-relative preorder over reporting commitments, not a global ordering of the four dependency statuses, and it remains subject to any qualifiers attached to the assessment. A system should not report stronger support than it assessed. But satisfying this reporting constraint does not establish that the assessment itself was correct:

$$\widetilde{\delta} \preceq_{\eta} \widehat{\delta} \quad \not\Rightarrow \quad \widehat{\delta} = \delta^{*}.$$

These are distinct transitions in the control chain. Runtime governance can directly constrain the relation between assessed and reported status, but it cannot thereby establish that the assessment matches reference dependency status; assessment remains subject to the same projection limits that motivate the framework as a whole.

Action or operation selection should be represented separately from these status variables and from the continuation prescribed by a compliant selection policy. Let

$$o_{\text{selected}} \in \mathcal{O}_{\eta}$$

denote the continuation actually selected by the system. Governance compliance requires that the selected continuation belong to the admissible set induced by the system's assessed dependency-status profile:

$$o_{\text{selected}} \in \mathcal{O}^{\text{adm}}_{\eta}\big(\widehat{\Delta}_{\Gamma,q,R}\big).$$

An *action-governance failure* occurs when this compliance condition fails:

$$o_{\text{selected}} \notin \mathcal{O}^{\text{adm}}_{\eta}\big(\widehat{\Delta}_{\Gamma,q,R}\big).$$

The violation can be localized further. Selection outside $\mathcal{O}_\eta^{\mathrm{ep}}$ is an epistemic-licensing failure; selection outside $\mathcal{O}_\eta^{\mathrm{valid}}$ is an operational-validity failure. Either places the selected continuation outside the admissible intersection.

A different case occurs when the system faithfully follows the operation policy induced by $\widehat{\Delta}_{\Gamma,q,R}$, but $\widehat{\Delta}_{\Gamma,q,R} \neq \Delta^*_{\Gamma,q,R}$. The system may then be governance-compliant relative to a mistaken structural assessment while still selecting an operation that would not be admissible under the actual dependency-status profile. Assessment correctness and governance compliance are therefore separate properties.

This yields a useful failure taxonomy. *Representation insufficiency* concerns the actual structural support available for the query. *Assessment failure* occurs when the bounded reasoner mischaracterizes that support. *Reporting failure* occurs when the reported epistemic status is stronger than the reasoner's own assessment warrants. *Action-governance failure* occurs when the selected operation or continuation violates the constraints associated with the assessed status. These failures may coexist, but they should not be treated as one undifferentiated category of hallucination or incorrectness.

The purpose of this section is not to specify a complete runtime protocol, evaluation system, or optimization policy. Its narrower goal is to show how the dependency-status interface yields operational targets: answer-form governance, evaluation under projection stress, interface-level representation repair, and audit fields for downstream systems.

The base notation records the assessed status class, while uncertainty about that classification may be represented through an attached qualifier rather than as another primitive status. When the assessment process records such uncertainty, the qualifier should remain visible in the reported status rather than disappearing through silent promotion. This requirement is analogous to Swanson's *non-laundering* condition for typed institutional reason-giving, under which reported grounds should track the grounds that actually did the decisional work [29]. In the present setting, uncertainty, assumptions, provenance limits, or unresolved dependencies that materially determine the reasoner's assessment should not disappear when the assessment is converted into a user-facing claim, action, or reasoning artifact.

7 Conditions for Structurally Reliable Correction

Binding dependency status to epistemic operation does not by itself ensure that the relevant status remains visible or correctly represented during later reasoning. A system may fail to expose that a required distinction is missing, silently import unsupported information during correction, or transfer support from one representational object or level to another. The framework therefore identifies three structural requirements for keeping dependency-status governance effective under projection: observability of loss-relevant state, bounded correction, and non-collapse of operationally distinct levels.

The requirements correspond to three different routes by which represented support can be distorted. Failure of observability allows consequential loss to disappear from the operative state. Failure of bounded correction allows unavailable distinctions to be silently introduced as though they had been recovered from the original representation. Failure of non-collapse allows support belonging to one object, role, level, or representation to be transferred to another without preserving the relation that licenses the transfer.

These requirements are conceptual design constraints rather than a complete axiomatization of reliable reasoning. The present paper does not claim that they exhaust every necessary condition or guarantee sufficiency across every regime. Nor are they substitutes for calibration, provenance, abstraction refinement, uncertainty representation, or compositional analysis. Their narrower role is to preserve the structural distinctions required for dependency status to continue governing epistemic operation.

7.1 Partial Observability of Loss-Relevant State

Observability of loss is necessarily partial and relative. A reasoning system need not be able to detect every distinction collapsed by projection; in general, it cannot. The requirement is not complete detection of loss, but preservation of detectable loss conditions relative to the representational language, metadata, or meta-representation available to the system. This includes:

- ambiguity between multiple consistent interpretations,
- absence of necessary variables or context,
- dependence on assumptions not encoded in the representation.

In the formal terms above, partial observability means that the system can sometimes detect evidence that the requested query q may not be constant over the equivalence class induced by P, relative to the distinctions it can express. When such evidence is available, the system must preserve it rather than treating the query as resolved.

Without observability, the system cannot distinguish between an identifiable query with a known answer and a non-identifiable query that only appears determinate after projection.

If such conditions are not observable within the system, they are effectively treated as resolved, even when they are not. This leads to outputs that appear complete but are based on unacknowledged gaps.

Observability does not require recovery of the missing information. In information-theoretic terms, it requires preserving enough side information about the projection, metadata, provenance, missing variables, ambiguity, or unsupported assumptions to represent detectable insufficiency as insufficiency. Failure of observability produces hidden loss: the system proceeds as if all relevant distinctions were present even when the representation no longer supports them.

7.2 Bounded Correction

A correction is bounded relative to a projection $P : S \to R$ when the transformation from the current result to the revised result is licensed entirely by distinctions represented in R, without an unmarked introduction of distinctions from outside R. Here R includes not only object-level data, but any explicitly represented metadata, provenance information, uncertainty markers, priors, task constraints, or assumptions available to the reasoning process. This definition does not require the reasoner to establish ground-truth incorrectness from R alone. It requires the basis of the revision to remain attributable to the operative representation. If revision requires distinguishing states that are equivalent under P and nowhere distinguished in R, then the process requires representation expansion, new information, or an explicit assumption that changes the represented information basis.

While bounded correction concerns preserving the information basis during revision, non-collapse concerns preserving the role distinctions among representational objects. A system may satisfy bounded correction yet still collapse levels by treating a representation as equivalent to the system it represents, or by treating a selected explanation as uniquely determined by the available evidence.

The important point is not that correction must be small in a purely computational sense. It is bounded when the correction is licensed by distinctions already encoded in R, or by an explicit refinement that produces a new representation R'. A correction that silently imports distinctions not encoded in R is not correction within R; it is an unmarked transition to a different representational state. For example, recomputing a formula from the same represented table is bounded correction; inventing a missing variable needed by the formula is not.

It is useful to distinguish two cases. In correction within R, the query is identifiable under the current representation and the system revises an incorrect answer using distinctions already available in R. In correction by refinement, the system explicitly moves from R to a richer or different representation R' by obtaining new information, adding a stated assumption, retrieving a source, or exposing metadata that was not previously represented. The second case may be admissible, but it is not correction within the original representation. It is bounded only if the transition from R to R' is explicit and the new information basis is marked. Thus bounded correction does not mean that a representation can never change. It means that correction must not silently change the information basis of the representation while presenting the result as if it followed from the original R.

A related consequence applies when reasoning under unresolved representational uncertainty must guide intervention rather than merely revise a claim. Bounded correction does not itself determine which action should be preferred, since action selection may depend on utilities, policies, authority, and other decision commitments not supplied by the represented evidence. It does, however, expose a structural difference among otherwise admissible actions. When the current representation leaves loss-relevant distinctions unresolved, an action that preserves observability, reversibility, and a bounded path for subsequent correction retains capabilities that an irreversible intervention may destroy. Thus uncertainty about the adequacy of the current representation can support a preference, under an explicitly declared decision rule, for interventions whose consequences remain detectable, attributable, and boundedly revisable. This is an interventional consequence of bounded correction rather than an extension of evidential support: it constrains how strongly a system commits under unresolved loss without determining the substantive value or policy according to which the action is selected [8].

This requirement separates valid correction from implicit reconstruction. If correcting a result requires unsupported new assumptions, wholesale replacement of the representation, or unbounded reconstruction of the problem state, then the issue is no longer ordinary error correction. It is either representational loss or a representation failure.

Bounded correction ensures that:

- revisions do not silently change the information basis of the representation;
- the distinction between correction within R and refinement to R' remains explicit; and
- external retrieval, intervention, or information acquisition is marked as a change of information basis rather than presented as support already contained in the original representation.

Failure of bounded correction leads to systems that appear to improve through repeated revision, but in fact introduce new assumptions or hidden state at each step.

7.3 Non-Collapse of Distinct Levels

For dependency-status governance to function as intended, a reasoning system must preserve role distinctions between objects that occupy different positions in the governed reasoning process. In dependency-status terms, non-collapse requires that support not be transferred across representational levels unless the dependency licensing that transfer is itself preserved, boundedly reconstructible, or explicitly conditioned. A representation may support a claim about the representation without supporting the same claim about the underlying system. A proxy may support a claim about the proxy without supporting the same claim about the target. A selected explanation may be consistent with the represented evidence without being uniquely licensed by it.

The relevant invariant is therefore not that all levels have the same structure, but that dependency status must not be silently promoted across levels. A preserved dependency at one level must not be treated as preserved at another level unless the relation between those levels is also preserved or reconstructible. Likewise, a traceable or opaque dependency must not become apparently reconstructed merely because it is attached to a fluent explanation, a metric, a provenance pointer, a selected model, or an aggregate representation.

This can be understood as a requirement of *typed support preservation*. Support is attached not only to a proposition, but to a relation between a particular reasoning object or level and the dependency that licenses a claim about it. A transfer of support from one object to another therefore requires a represented transfer condition.

Schematically,

$$\mathrm{Supp}(x, d) \not\Rightarrow \mathrm{Supp}(y, d)$$

unless the relation carrying support from x to y is itself preserved, boundedly reconstructible, or explicitly conditioned. Non-collapse therefore prohibits *unlicensed support transfer*: a representation cannot inherit the epistemic status of the world, a proxy that of its target, a provenance pointer that of source content, or a local judgment that of a composed system merely because the two objects are operationally connected.

In particular, the system must not collapse:

- a representation with the underlying system it represents,
- a proxy or metric with the target property it measures,
- a provenance pointer with recoverable source support,
- provenance that a source participated in a determination with permission for that source to determine the target in the role it played,
- an intermediate reasoning state with a final conclusion,
- a selected explanation with the full set of explanations consistent with the representation,
- persistence of a reasoning operation with preservation of the conditions that license its use,
- a local component-level judgment with a claim about the composed system.

Determination provenance should therefore remain distinct from determination permission. A representation may correctly preserve which source or commitment selected an outcome while still losing the source–target–role constraint needed to establish that the represented determination was licensed. Knowing what determined a result does not, by itself, establish that it was permitted to determine that target in that role.

The same distinction applies to reusable reasoning operations. A representation may preserve enough structure to execute a heuristic, procedure, scaffold, or transformation without preserving the context in which that operation was originally justified. Executability is therefore a property of the represented operator, while applicability is a dependency claim about the relation between that operator and the present state. Collapsing these levels allows a successfully preserved method to acquire unwarranted authority merely because it remains usable.

In each case, the collapse has the same form: a projected object inherits a stronger dependency status than the representation licenses. A representation is treated as the world, a proxy as the target, a pointer as source support, a partial reasoning state as a conclusion, a selected explanation as uniquely determined, or a local component judgment as a composed-system judgment. This makes projection loss invisible because the system can no longer track which dependency status belongs to which level.

These distinctions may coincide in specific cases, but they are not equivalent in general. Collapsing them prematurely removes the ability to track where information originates and how conclusions are derived.

For example, treating a selected explanation as if it were uniquely determined by the data conflates the space of possible models with the subset supported by the representation. Similarly, reducing a multi-dimensional decision to a single value collapses distinct evaluative dimensions into a single outcome.

Failure of level separation results in premature convergence, where the system commits to a single interpretation without preserving alternative possibilities that remain consistent with the available information.

Relational structure and arity. The framework frequently represents important distinctions in paired form, but these pairs should not all be interpreted as the same kind of duality. Some represent independently necessary dimensions of admissibility, some represent operational roles that must remain distinguishable, some compare alternative transformation paths, and some are simply the smallest nontrivial instance of a more general relation among several represented states or reasoners.

For example, query-relative support and operational validity are distinct licensing questions:

$$\text{informational support} \neq \text{operational permission.}$$

Evidence, background assumptions, and policy constraints are instead distinct support roles within one operative information basis. Target formation and representational projection are distinct transformations in a composed process. In other applications of the framework, two reasoners or two reasoning bundles may be used as the smallest illustrative case even though the relevant structure extends to

$$R_1, \ldots, R_n.$$

The common principle is therefore not binary structure as such. It is *relational non-collapse*: when a downstream continuation depends on the distinction among states, roles, transformations, or participants, or on a material relation among them, projection should preserve or boundedly reconstruct that distinction and relation rather than silently replacing them with an undifferentiated state.

Accordingly, the use of two components in an example should be understood as illustrative unless the relevant result specifically depends on a binary structure. Reconstructibility is query-relative to the distinctions and relations that matter, not to a fixed arity of the system being represented.

7.4 Structural Role of the Requirements

Together, these requirements preserve the distinction between what is known, what can be corrected, and what has been lost. A system that satisfies them does not eliminate failure; it keeps failures detectable and keeps reasoning grounded in the structure of the representation. When they are absent, outputs may appear coherent while the reasoning process has diverged from the information available to it.

8 Diagnostic Examples

These cases are not intended to estimate the frequency of projection-induced failures. They are diagnostic probes showing how the framework distinguishes unsupported determinacy, admissible conditional action, ordinary error, excessive abstention, and ambiguous dependency-status classification.

8.1 Case 1: Scalarization and Proxy Collapse

In this scenario, the system is required to produce an approval decision using a single numeric metric with no caveats. This constraint forces multiple dimensions of risk into a single value, creating the familiar possibility that optimization or decision-making over a proxy ceases to preserve the distinctions that matter to the underlying objective [11]. The relevant query is not merely "which option maximizes the metric?" but "which decision is justified given reversibility, uncertainty, causal structure, and decision criteria?" If two underlying states map to the same scalar value while requiring different decisions along one of these dimensions, then the decision query is not identifiable under the scalar projection.

A structurally reliable response should identify the scalar projection as insufficient for the decision query. The result need not be refusal. The system may request a richer representation, branch over unresolved risk dimensions, or make an explicitly labeled policy decision under non-identifiability. The inadmissible move is to present the scalarized answer as an inference justified by the metric alone.

8.2 Case 2: Responsibility Assignment Under Causal Collapse

In this scenario, the system is asked to assign responsibility for a failure to a single individual, despite the presence of multiple interacting causes. The relevant query is not simply "who can be named as responsible?" but "which causal structure best explains the failure?" If distinct causal structures map to the same single-person attribution, then responsibility is not identifiable under that projection.

A structurally reliable response should avoid treating one selected actor as if that actor were uniquely determined by the representation. It may instead separate individual action, process failure, system-level interaction, and unresolved evidence. The point is not to avoid responsibility claims altogether, but to preserve the dependency status of the attribution being made.

8.3 Case 3: Traceability Without Source Reconstruction

A common language-model failure occurs when a system is asked to summarize or cite information that is not present in its context. The projected representation contains topic cues, stylistic patterns, and plausible associations, but it does not contain the source-level distinction between what was actually observed and what is merely likely.

The relevant query is not simply "what answer is plausible?" but "which claims are supported by the provided source?" If two underlying states map to the same representation—one in which a claim is source-supported and one in which it is merely plausible—then source support is not identifiable under the projection. A structurally reliable response should separate supported claims from plausible but unsupported claims, request the missing source, or decline to assert source support.

8.4 Case 4: Decision Under Non-Identifiability

Not every non-identifiable query requires inaction. Suppose a triage system must choose between two actions under time pressure, but the representation does not contain enough information to determine which action is uniquely optimal. Two underlying states may map to the same representation while requiring different optimal actions under full information.

In this case, the query "which action is uniquely best?" is not identifiable under the projection. However, a decision may still be justified by an explicit rule, such as minimizing worst-case harm, deferring actions that would commit the system to loss beyond bounded correction, or selecting the option with a bounded fallback path. The decision is admissible only when its license depends explicitly on a policy choice under non-identifiability and the resulting action is not represented as uniquely evidence-determined.

The chosen policy must also satisfy the operational conditions of the declared regime. Marking an action as a decision rule resolves the inference-status problem, but it does not by itself establish that the action is authorized, reversible, feasible, or otherwise operationally valid.

8.5 Case 5: Ambiguous Dependency Status Under Limited Observability

Some cases cannot be cleanly classified from the current representation alone. Suppose a system gives an unsupported causal explanation, but the representation does not show whether the explanation came from hidden evidence, a valid prior, or an invented assumption. From the output alone, the failure may look like representational loss, ordinary error, or unjustified decision under uncertainty.

In this case, the framework does not by itself determine the correct classification. It identifies what must be made observable: whether the explanation is supported by information in the representation, by an explicit prior or decision rule, or by an unsupported assumption. This limit case is important because the framework does not eliminate projection; it specifies what additional structure must be exposed for classification to be reliable.

8.6 Summary of Behavioral Modes

The examples distinguish restructuring, conditional decision, representation expansion, trace marking, and refusal of the requested answer form as possible responses to non-identifiability. They also show that the framework should not be used as a blanket abstention policy: when a query is identifiable under the current representation, ordinary correction remains appropriate. The central requirement is not caution for its own sake, but matching the answer form to the dependency status of the claim.

9 Applications

Dependency status can be used as a design and evaluation lens for several AI reasoning, safety, and security problems.

Security copilots, code assistants, and incident response. Security and incident-response assistants often reason from compressed code context, alert summaries, deployment traces, and partial telemetry. The framework requires such systems to distinguish vulnerabilities or root causes supported by represented evidence from recommendations that depend on missing code paths, runtime configuration, threat models, event ordering, or logging conditions. When these distinctions are absent or only traceable, the system should mark the missing dependency, propose discriminating tests, request representation expansion, or offer bounded mitigations rather than present a single unsupported cause or fix as determined.

Retrieval-augmented generation and source grounding. RAG systems frequently expose provenance pointers, citations, or retrieved snippets. The dependency-status ladder distinguishes traceability from reconstructibility: a pointer records a dependency, but only available source content and recovery rules can support the claim. This provides a concrete audit target for hallucination and false-grounding failures.

Agentic systems. Consider an agent asked to delete a production database. The proposed action may depend on at least four distinct conditions:

$$D_{\text{delete}} = \{d_{\text{identity}}, d_{\text{authorization}}, d_{\text{state}}, d_{\text{reversibility}}\}.$$

An illustrative actual dependency-status profile might be

$$\Delta^*_{\Gamma,q,R}(D_{\text{delete}}) = \{d_{\text{identity}} : \mathsf{P},\ d_{\text{authorization}} : \mathsf{T},\ d_{\text{state}} : \mathsf{R},\ d_{\text{reversibility}} : \mathsf{O}\}.$$

Here the database identity is directly represented, the authorization dependency is attributable to an approval record but has not been recovered, the current database state can be reconstructed by a permitted bounded check, and no adequate representation of reversibility or rollback remains. Direct deletion is not thereby admissible. Preserved target identity does not supply authorization, and successful execution would not supply reversibility after the fact. Traceable authorization may instead license retrieval or routing for approval; reconstructible state may license the bounded check; and opaque reversibility may require a backup, representation expansion, or refusal of the destructive action.

The example illustrates why an agent should retain a dependency profile rather than one overall confidence value. Different dependencies can license different preparatory operations, while the associated action remains subject to their joint epistemic and operational requirements. More generally, agents acting from compressed instructions, memory, tool outputs, environmental observations, and inferred user intent should not treat evidence about an action's target as evidence that the action is authorized, safe, or reversible.

User modeling and Theory-of-Mind-like behavior. Dependency status also applies to AI systems that model users or other agents. A system often infers what a user knows, wants, intends, believes, or cannot distinguish from partial interaction traces. These inferences are projection-sensitive: observed behavior may preserve some distinctions while leaving others merely plausible, traceable to prior context, or opaque. A safety-relevant user model should therefore mark whether claims about another agent's state are directly observed, reconstructible from interaction, dependent on an explicit assumption, or unsupported. This does not require treating the AI system as having humanlike inner experience. It treats Theory-of-Mind-like behavior as reasoning under projection over another agent's representation.

Reasoning scaffolds, procedures, and inherited decision rules. Reusable reasoning procedures are themselves subject to projection. A prompt scaffold, diagnostic checklist, review procedure, policy rule, or institutional heuristic may remain available long after some of the assumptions or contextual distinctions that originally governed its use have disappeared from the operative representation. A system should therefore distinguish whether it can execute a procedure from whether the dependencies licensing that procedure remain preserved or reconstructible in the current case. Successful execution does not by itself establish appropriate application.

Evaluation and benchmark design. Standard evaluation may reward correct-looking answers even when the representation does not support them. Projection-aware evaluation should test whether systems change behavior when observability changes: refusing unsupported determinacy at low observability, ranking hypotheses under partial evidence, and becoming appropriately decisive when the relevant distinction becomes observable. This makes dependency status a measurable target rather than a purely philosophical concern.

These applications are not offered as evidence that the underlying domains share one ontology or that their established methods can be replaced by a single taxonomy. Their commonality is narrower and operational. Each contains a boundary at which a projected representation supplies support for a downstream claim, procedure, or action; each therefore requires some way to distinguish direct availability, bounded recovery, attributable but unavailable support, and opacity. Dependency status functions as a thin waist precisely because the representations above it and the workflows below it may remain domain-specific while sharing a small vocabulary for preventing unsupported promotion across that boundary.

10 Why Existing Methods Can Miss This Failure

Existing methods can improve performance within a representation without testing whether the representation preserves the distinctions required by the query. Optimization can improve a proxy while hiding loss of task-relevant structure. Prompting can make reasoning more detailed without verifying that the needed dependency is represented. Evaluation can reward a plausible answer while failing to test whether a determinate answer was identifiable from the available evidence.

The common mismatch is that these methods often operate over R, while the failure arises from the projection $P : S \to R$. A source-grounding benchmark may check whether an answer contains a citation, but not whether the cited content preserves the distinction needed by the claim. An incident-response benchmark may reward naming a plausible cause, while failing to penalize the system for ignoring that telemetry cannot distinguish among several causes.

This does not mean optimization, prompting, or evaluation are misguided. It means they require an additional structural check: whether the claim's dependency status is preserved, reconstructible, traceable, or opaque under the representation being used, and whether a proposed continuation satisfies the operational-validity conditions of the declared regime or preserves an explicitly permitted path for establishing them.

11 Relation to Existing Frameworks

The framework draws on several established treatments of reasoning under incomplete or reduced information. Table 3 summarizes the closest comparisons.

Across these traditions, the closest common point is query-relative sufficiency. The contribution here is not another general result that compression can destroy task-relevant information. It is an operational interface for what follows: dependency status records whether required support is directly preserved, boundedly reconstructible, attributable but unavailable, or opaque, and binds those conditions to different epistemic operations and response forms.

Bruhacs provides an especially close exact-sufficiency baseline by characterizing the invariants, selectors, and operators that must survive projection for a task, together with limits on task-blind reconstruction across heterogeneous targets [4, 2, 3]. The present framework addresses the bounded-reasoner regime in which that threshold is not met or cannot be established. It asks what continuation remains licensed rather than supplying another criterion for exact recovery.

Swanson's distinction between target formation and subsequent rendering, and the associated analysis of bounded carriers, helps separate what was excluded from the modeled target from what was lost in its operative representation [28, 27]. The present account uses that distinction operationally: different repairs are required for an inadequate target boundary and an inadequate projection, and neither informational repair by itself establishes operational permission.

POMDP-style methods represent uncertainty over modeled hidden states, but hidden state is not identical to projected-away or unrepresented structure. A belief state can regulate action among possibilities the model contains without establishing that every query-relevant distinction survived the construction of that model. Dependency-status governance is therefore complementary: it asks whether the distinction needed by a proposed continuation is available, recoverable, traceable, or absent even as an accountable dependency.

The separation between representation, assessment, and output also has empirical motivation. Slobodkin et al. find that language models may encode information predictive of question answerability while still producing hallucinatory answers [25]. This does not establish the present theory, but it illustrates why represented support, assessed support, reported support, and selected behavior should be evaluated separately.

The unification claimed here is consequently operational rather than ontological. The neighboring fields retain their own objects, semantics, and methods. They meet at a narrower question:

What operation and response form remain licensed by the dependency structure that survived projection?

12 Operational and Cybernetic Implications

A minimal implementation can treat admissibility assessment as a five-step diagnostic rather than as an oracle:

1. Identify the claim, action, or continuation being licensed.
2. Identify the distinctions on which that claim, action, or continuation depends.
3. Check whether each distinction is preserved, reconstructible by a bounded procedure, merely traceable, or opaque in the operative representation.

Table 3: Neighboring frameworks and the additional operational question posed here.

Framework	**Principal question**	**Additional question in this paper**
Identifiability, partial identification, and certain answers	What values or claims are determined across observationally equivalent states or admissible completions? [21, 19, 20, 17]	When unique determination fails, is the required dependency directly available, boundedly recoverable, merely attributable, or opaque, and what response form does that condition license?
Blackwell comparison, task sufficiency, and information bottlenecks	Which representations preserve the information required for a task or decision family? [1, 30, 4, 2]	What operation remains licensed when the operative representation falls below, or cannot be shown to meet, that sufficiency threshold?
State abstraction, abstract interpretation, and refinement	Which properties, transitions, policies, or values survive an abstraction, and how should an inadequate abstraction be refined? [16, 6, 5]	How should a bounded reasoner classify surviving support and govern its current answer or action before refinement succeeds?
Partial observation and epistemic awareness	How should an agent reason over hidden state, and which propositions or concepts are available to it? [13, 7]	Does the representation preserve an accountable dependency even when its content is unavailable, and which continuation does traceability license that opacity does not?
Reconstructability and provenance	Can system structure be recovered from component relations, and can the origins or transformations of an artifact be followed? [14, 15, 26]	Does the preserved relation recover the query-relevant distinction, or does it provide attribution without reconstruction?
Selective prediction, conformal prediction, and hallucination detection	When should a predictor abstain, return a prediction set, or flag unsupported generation? [9, 10, 31, 23, 32, 18]	Is the answer form licensed by dependency structure independently of the system's confidence or realized correctness?
Representation-risk and boundary accounts	Where was a target or representation narrowed, and what task loss does that bounded representation induce? [28, 27]	How should the remaining dependency status govern inference, reconstruction, qualification, routing, or action?

4. For a proposed action or continuation, check whether it satisfies the operational-validity conditions of the declared regime or preserves an explicitly permitted path for establishing them.

5. Select an epistemic operation and associated response form jointly licensed by dependency status and operational validity: direct inference, bounded reconstruction, conditionalization, branching over assumptions, representation expansion, retrieval, trace marking, routing, constraining, deferral, or refusal of the requested form.

The distinctions among actual, assessed, and reported dependency status and between prescribed and actually selected operations, together with their corresponding failure taxonomy, are developed in Section 6.2. The remaining subsections show how those distinctions support evaluation, reasoning scaffolds, interface design, question formation, and audit.

12.1 Dependency Status in a Corrective Loop

The framework has a cybernetic interpretation, but dependency status should not be confused with a complete controller state. A bounded reasoner receives or constructs an operative representation, assesses the dependencies required by a proposed continuation, constructs the admissible continuation set, and applies separate budget and policy considerations to prescribe a continuation from that set. Controller execution then produces the continuation actually selected, which may comply with or depart from the prescription and admissible set. The resulting compliance assessment and observed consequences may reveal ordinary error, expose an unresolved dependency, identify an action-governance failure, or create diagnostic pressure to refine the representation.

Schematically,

$$R_t \longrightarrow \widehat{\Delta}_t \longrightarrow \mathcal{O}_t^{\mathrm{adm}} \longrightarrow o_{\mathrm{prescribed},t} \longrightarrow o_{\mathrm{selected},t} \longrightarrow y_t \longrightarrow R_{t+1}.$$

Here R_t is the operative representation, $\widehat{\Delta}_t$ is the assessed dependency-status profile, $\mathcal{O}_t^{\mathrm{adm}}$ is the admissible continuation set, $o_{\mathrm{prescribed},t}$ is the continuation prescribed by selection and budget policy, $o_{\mathrm{selected},t}$ is the continuation actually selected by the system, and y_t contains the compliance assessment, observed consequence, or other feedback that becomes representationally available for later reasoning. The transition from y_t to R_{t+1} may leave the representation unchanged, correct an error within it, retrieve or reconstruct missing support, record a governance violation, or refine the represented dependency structure.

This loop is corrigible rather than self-certifying. Successful execution or favorable feedback does not establish that $\widehat{\Delta}_t = \Delta_t^*$, that every material dependency was represented, or that the same compression remains adequate for a later query or recipient. Feedback can improve control only through distinctions that remain observable or become observable through representation change. Dependency status therefore supplies a compact control interface for prescribing structurally admissible continuations and evaluating actual selection without claiming that the interface contains the entire state required for optimal control.

12.2 Evaluation by Structural Shape

Evaluation can be organized around whether a response preserves the structural shape required by the query, rather than only whether it matches a domain-specific target answer. A responsibility query can be evaluated for preservation of individual, process, and system levels; a high-stakes decision for visibility of irreversible tradeoffs and residual risk; a source-grounded answer for separation between observed support and plausible reconstruction; and a compositional policy problem for revalidation under interaction or scale. Such diagnostics ask whether the answer has the right admissibility structure: what distinction is needed, whether it is preserved or reconstructible, whether loss is visible, and whether the system has confused inference with decision. These targets can be operationalized through controlled question sets that vary the operative representation, prompt conditions, and reasoning scaffold while holding the underlying task fixed.

The separation among actual, assessed, and reported status, policy prescription, and actual selection also suggests that evaluation should localize failure rather than reduce performance to answer accuracy. The corresponding evaluation measures use performance rather than failure terminology: assessment failure is measured by assessment accuracy, reporting failure by reporting fidelity, action-governance failure by governance compliance, and the combined effect of representation and downstream behavior by end-to-end structural compliance. Assessment accuracy asks whether $\widehat{\Delta}_{\Gamma,q,R} = \Delta^*_{\Gamma,q,R}$, or, more finely, which individual dependency statuses were misclassified. Reporting fidelity asks whether the communicated epistemic status is no stronger than the assessed profile permits. Governance compliance asks whether $o_{\text{selected}} \in \mathcal{O}^{\text{adm}}_{\eta}(\widehat{\Delta}_{\Gamma,q,R})$. End-to-end structural compliance asks whether the selected continuation would be admissible under the actual profile, $o_{\text{selected}} \in \mathcal{O}^{\text{adm}}_{\eta}(\Delta^*_{\Gamma,q,R})$. The epistemic and operational components can also be scored separately to distinguish a support-licensing violation from an operational-validity violation. These measurements distinguish a system that faithfully acts on a mistaken structural assessment from one that correctly assesses insufficiency and nevertheless violates the corresponding response constraint.

12.3 Reasoning Scaffolds and Constraint Activation

The same dependency-status distinctions can be used to constrain generation. A reasoning scaffold can require the system to preserve observability of loss-relevant state, avoid collapse of conceptual levels, keep correction bounded, distinguish error from loss, and revalidate conclusions under scale or composition. Such scaffolds do not prescribe a single reasoning style. They constrain which transformations are admissible when the system moves from a projected representation to an answer.

In addition, different tasks may require attention to different dependency-status risks. A source-grounding task may foreground observability and provenance; a responsibility-assignment task may foreground non-collapse of levels; a deployment decision may foreground bounded correction and scale revalidation. Task-specific activation of these dependency-status risks provides a control layer over reasoning without changing the underlying structural requirements.

12.4 Interface and Question-Formation Uses

Dependency-status distinctions can also be used before answer generation, as interaction lenses for forming better questions. A user interface need not treat the query as fixed. It can help users select objects, group them, specify relations, and choose a reasoning lens such as observability, reconstructibility, traceability, admissibility, scale, or non-collapse. The resulting query is then not merely a request for an answer, but an explicit statement of which distinctions must be preserved during reasoning.

On this view, question formation is itself a projection-management task. The user and system jointly determine which distinctions matter before inference proceeds. This is especially important in settings where the initial question collapses roles, scales, sources, or decision criteria that must remain separate for the answer to be admissible.

In interactive systems, clarifying questions are one mechanism for representation repair. A clarification request is not merely conversational hesitation or a user-interface convenience. It is a way of enriching, refining, or changing the projection so that the requested query becomes identifiable, boundedly reconstructible, or explicitly conditional. Conversely, if the missing distinction does not affect the dependency status of the requested claim, the system should not treat clarification as required. The point is not to ask more questions, but to ask when the current representation does not support the requested answer form.

Question formation can also reveal dependencies that were absent from the initial problem model. A clarification may split one overloaded query into several queries, expose a missing stakeholder, distinguish evidence from authority, or make an existing concern newly relevant to a downstream action. Such events do more than enrich the current representation with additional facts. They revise the structure to which admissibility is applied. Call this process *constraint discovery and contextual persistence*: newly discovered relationships should remain attached to the claims, actions, questions, or projections they limit until their disposition is made explicit.

12.5 Audit and Dependency Status

Finally, the dependency-status ladder suggests audit fields for downstream systems. A claim can be marked according to whether its required dependency is preserved, reconstructible, traceable, or opaque. Such markings do not guarantee correctness, but they make visible whether a conclusion is supported by the representation, recovered through an explicit path, dependent on a recorded but unrecoverable relation, or unsupported by observable provenance.

This allows downstream users or systems to distinguish claims that are inferentially supported from claims that are merely decisions, assumptions, or unresolved dependencies under non-identifiability.

Taken together, these uses show how dependency status can guide downstream scaffolds, evaluation harnesses, runtime systems, interfaces, and audit mechanisms without requiring a single domain-specific answer metric.

13 Scope and Limits

This paper describes structural conditions for reasoning under projection. It does not attempt to provide a complete theory of reasoning, nor does it claim to resolve all sources of uncertainty or failure. This section clarifies the scope of the claims and the limits of what these constraints guarantee.

13.1 What This Does Not Solve

Respecting the constraints described in this paper does not eliminate the fundamental challenges associated with reasoning over incomplete or compressed representations.

In particular, it does not:

- guarantee that a system will produce correct conclusions,
- eliminate ambiguity in cases where multiple interpretations remain consistent,
- recover distinctions that are not preserved or boundedly recoverable under the current representation,

- ensure that all relevant variables are present in the representation,
- prevent all forms of bias or misalignment in decision-making,
- guarantee that the system has discovered every dependency, stakeholder, constraint, alternative method, or intended use relevant to the query.

These limitations are not defects of the account. They follow from projection and from the developmental incompleteness of practical reasoning. A system may faithfully govern the dependency structure it currently represents while still failing to notice a material relationship. Newly discovered constraints can revise that structure and should continue to govern later reasoning while they remain materially operative and legitimately retained. This does not imply that discovery can ever be complete or that discovered state must be retained permanently. A system may likewise preserve and correctly execute a reasoning procedure while failing to represent a condition that limits when that procedure is applicable.

13.2 Dependence on Representation

The behavior of a reasoning system under this framework depends on the representation over which it operates. Different representations preserve different distinctions, and therefore introduce different forms of loss.

Improving reasoning performance may require modifying the representation itself, not just the reasoning process applied to it. However, such changes alter the projection and therefore change the conditions under which reasoning operates.

This paper does not prescribe a single representation or assume that the operative representation remains fixed. It provides constraints that apply to a representation at a given stage of reasoning. A later stage may refine the projection, introduce a new distinction, attach a newly relevant constraint, or reformulate the query. Such revisions alter the state to which admissibility is applied and should therefore be explicit rather than silently attributed to the earlier representation.

Representation enrichment does not in general imply monotonic expansion of the admissible continuation set. In particular,

$$R' \succeq_q R \quad \not\Rightarrow \quad \mathcal{O}^{\mathrm{adm}}_{\Gamma,q,R}(\Delta) \subseteq \mathcal{O}^{\mathrm{adm}}_{\Gamma,q,R'}(\Delta').$$

A richer representation may reveal a previously omitted dependency, contradiction, authority limitation, safety condition, or interaction that removes a continuation from the admissible set. Licensing monotonicity can be expected only under fixed material dependencies, operational-validity conditions, and regime, together with a conservative refinement that adds query-relevant distinctions without invalidating previously represented support.

13.3 Transport, Persistence, and Executability

Preserving or transporting a reasoning artifact does not by itself preserve its adequacy for a later query. A persistent bundle, versioned object, provenance record, memory item, or transferred dependency profile may retain information that would otherwise have been lost, but its arrival in a new context does not establish that it preserves every distinction required there. State transport preserves an artifact and whatever represented relations survive the transport; it does not certify destination-relative reconstructibility or admissibility.

The same limitation applies to executable procedures and tooling. A procedure may remain available, syntactically valid, and successfully executable after the assumptions, authority, evidence, or environmental conditions that licensed its earlier use have changed or disappeared. Persistence can improve traceability and create bounded recovery paths, but neither persistence nor executability supplies the missing applicability conditions by itself. Recipient systems must therefore reassess dependency status and operational validity relative to their own query, regime, operation repertoire, and operative representation.

13.4 Diagnostic Pressure and Earned Observability

The fact that a representation may omit a materially relevant distinction does not imply that every conceivable distinction should be represented in advance. Such a requirement would replace one failure mode with an unbounded demand for representational completeness, which the framework neither assumes nor requires. Instead, refinement can be driven by *diagnostic pressure*: evidence that the current representation repeatedly prevents a materially relevant failure, disagreement, ambiguity, or unresolved alternative from being localized.

Diagnostic pressure may arise when independent reasoning processes diverge at an unresolved dependency, when repeated failures remain observationally indistinguishable, when an investigation repeatedly encounters inadequate provenance or logging, when correction requires information that the current representation consistently fails to preserve, or when an observed failure does not fit the distinctions supplied by the current model. The last case is especially important in adversarial and open-ended environments: the relevant failure class may not have been known when the representation was designed. Such evidence does not automatically identify a missing variable, dependency, or ontology, nor does it require permanent expansion of the modeled state. It establishes a narrower claim: the current representation is insufficient for diagnosing something that practical experience has shown to matter.

The resulting refinement should therefore be minimal and diagnostic where possible. A system may add telemetry, provenance, an explicit dependency relation, a discriminating test, a temporary probe, a retained distinction, or a richer representation. The aim is not to represent more state for its own sake, but to make a demonstrated class of failures more observable, attributable, or correctable. In this sense, additional structure *earns* its place in the representation through demonstrated diagnostic value.

The same principle applies in reverse. Structure need not be retained merely because it was once represented. Dependencies that are redundant, cheaply reconstructible, no longer material to the governed query class, or costly to preserve relative to their diagnostic value may be compressed, summarized, or removed. Such dependency hygiene is compatible with structural reliability so long as the resulting representation does not silently claim support that depended on the discarded structure. The objective is sufficient diagnostic capacity, not maximal retention. Removal may also be required independently of diagnostic value by privacy, legal, security, contractual, or other governance constraints; in that case the structural requirement is not continued retention but correct reassessment of any surviving claim or operation whose support depended on the removed state.

13.5 Implementation and Evaluation Boundaries

The present paper specifies the structural failure mode rather than a complete implementation. A detector for representational loss would need to identify when a query depends on distinctions that are neither preserved, nor boundedly reconstructible, nor explicitly assumed in the current representation. A dependency-status implementation would need to represent whether a claim is preserved, reconstructible, traceable, or opaque.

Evaluation should therefore test not only whether a system reaches a target answer, but whether it changes behavior appropriately as the representation changes. One natural benchmark pattern is to compare compressed and enriched versions of the same task: when the compressed representation does not preserve the distinction required by the query, the system should avoid unsupported determinacy; when the enriched representation restores the distinction, the system should become correspondingly more decisive. This tests whether the system distinguishes representational loss from ordinary error rather than merely becoming more cautious.

The present account therefore separates a structural theory from its possible implementations and evaluations. It specifies what a projection-sensitive system must distinguish and identifies behavioral changes that should follow from those distinctions, but it does not supply a complete runtime architecture, a universal response policy, or a validated benchmark. Those remain independent engineering and empirical questions rather than consequences of the formal definitions alone.

13.6 Empirical Vulnerability of the Practical Account

The formal claims in this paper are conditional on their stated representational assumptions. In particular, the non-identifiability result follows when distinct states collapsed by the operative projection require different answers to the query. A separate empirical question is whether the dependency distinctions developed here provide useful explanations or predictions for actual reasoning systems. That broader practical claim should remain vulnerable to evidence rather than being inferred from the internal consistency of the formal account.

Evidence against the practical importance of the framework would include failure to observe meaningful behavioral differences between preserved, reconstructible, traceable, and opaque dependency conditions under independently designed evaluation; failure of controlled representation enrichment to produce the predicted change in licensed answer form; failure of relational or dependency structure to add diagnostic value beyond component-level information; or repeated external cases in which the framework's distinctions provide no useful explanatory or corrective advantage over simpler existing accounts. Such findings would not invalidate the elementary non-identifiability result itself, but they would weaken the claim that reconstructibility under projection identifies a practically important structure of AI reasoning failure.

13.7 Self-Application Non-Certification

The framework is itself subject to the conditions it describes. An admissibility procedure operates over a representation of the dependencies, regime, query, and available recovery paths. It therefore cannot infer from successful execution alone that those represented conditions were complete or adequate.

This yields a general principle:

$$\boxed{\begin{array}{c}\text{successful execution of a reasoning or governance procedure}\\ \not\Rightarrow\ \text{certification that its application was licensed.}\end{array}}$$

Call this *Self-Application Non-Certification.* This is a representation-relative limitation principle, not an independent impossibility theorem. The principle does not make admissibility assessment futile. It limits what such assessment certifies. A procedure may improve structural reliability, expose detectable loss, constrain reporting failure, and require explicit recovery paths without becoming an oracle for the completeness of its own representation.

The practical consequence is that admissibility mechanisms should expose the basis and limits of their own assessments. They may classify represented dependencies, record unresolved conditions, request refinement, or qualify their own status judgments. They should not silently convert successful self-application into evidence that no relevant dependency was omitted upstream.

13.8 Limits of Observability

While the error–loss distinction requires observability of loss-relevant state, this observability is itself subject to projection. A system may be unable to represent certain forms of loss because the representation does not include the necessary structure.

This creates an important limitation: loss can only be detected relative to distinctions that the representation is capable of expressing. A system cannot observe every loss introduced by its own projection; it can only mark the boundaries where the current representation no longer supports a requested query.

In such cases, the system may be unable to indicate the specific lost distinction, even if loss exists. It may still be able to mark a weaker diagnostic condition, such as missing variables, unsupported assumptions, or insufficient provenance. This places a practical limit on how completely loss can be made visible within any given representation.

The practical target is therefore improved diagnostic visibility, not complete loss detection or representational completeness. The framework seeks to make consequential insufficiency increasingly observable where experience provides evidence that the current representation is inadequate, while retaining the recursive limitation that no represented diagnostic process can certify that every material dependency has been found.

14 Conclusion

Projection-induced non-identifiability is familiar. The contribution of this paper is the operational structure placed after it:

$$\boxed{\text{projection} \to \text{dependency status} \to \text{licensed epistemic operation} \to \text{response or action.}}$$

Preserved, reconstructible, traceable, and opaque status arise from a gated recovery–attribution construction that can equivalently be expressed through three nested capability thresholds. Under the declared surjectivity, status-sufficiency, and fiber-separation assumptions, they form the primitive behavioral quotient induced by the declared base licensing relation; threshold-witnessing operations provide a concrete sufficient condition for the required separation. Across a status-separating context family, no proper quotient preserves the same portable licensing behavior. This is a claim about a minimal operational interface relative to declared primitives, operations, and contexts, not a universal ontology or a requirement that every local controller maintain four literal internal states.

The central failure is unsupported promotion: traceability becomes reconstruction, policy becomes evidence, executability becomes applicability, or locally adequate support becomes portable authority without a represented witness for the stronger role. Preventing such promotion requires epistemic licensing, operational validity, admissibility, policy prescription, and actual selection to remain distinct, together with the actual, assessed, and reported dependency status on which those stages operate.

Dependency status can consequently function as a bounded cybernetic control variable. It allows a reasoner to change its operation as its information basis changes and enables observed failures to motivate targeted representation refinement. It does not guarantee that every material dependency has been discovered, that assessed status is correct, or that a governance procedure can certify its own applicability. Structural reliability under projection is therefore not omniscience. It is the more attainable discipline of keeping unsupported continuation visible, attributable, and correctable where the operative representation permits.

References

[1] David Blackwell. Equivalent comparisons of experiments. The Annals of Mathematical Statistics, 1953.

[2] Lorand Bruhacs. Hidden Structure and Explanation: Explanatory Descent and Minimal Hidden Support. Preprint, 2026. doi:10.5281/zenodo.19057860.

[3] Lorand Bruhacs. The Limits of AI Inference: Abduction, Regularization and Non-Public Norm Selection. Preprint, 2026. doi:10.5281/zenodo.19411423.

[4] Lorand Bruhacs. Task-Sufficient Inference and Explanation: Reasoning Under Projection. Draft preprint, 2026.

[5] Edmund M. Clarke, Orna Grumberg, Somesh Jha, Yuan Lu, and Helmut Veith. Counterexample-guided abstraction refinement. CAV, 2000.

[6] Patrick Cousot and Radhia Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. POPL, 1977.

[7] Ronald Fagin and Joseph Y. Halpern. Belief, Awareness, and Limited Reasoning. *Artificial Intelligence*, 34(1):39–76, 1987.

[8] Tim Fist, Saif Khan, Tao Burga, Arthur Tellis, Ben Schifman, Jonah Weinbaum, and Olivia Scharfman. How Should the US Prepare for Increasingly Automated AI R&D? 23 Low-Regret Policy Recommendations. Institute for Progress, August 6, 2026.

[9] Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, 2017.

[10] Yonatan Geifman and Ran El-Yaniv. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. In *Proceedings of the 36th International Conference on Machine Learning*, 2019.

[11] Charles Goodhart. Problems of Monetary Management: The U.K. Experience. In *Papers in Monetary Economics*, 1975.

[12] Joseph M. Hellerstein. Determination Provenance: From Ambiguity to Algebra. arXiv:2606.10270, 2026.

[13] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. Artificial Intelligence, 1998.

[14] George J. Klir and Eileen C. Way. Reconstructability Analysis: Aims, Results, Open Problems. *Systems Research*, 2:141–163, 1985.

[15] George J. Klir. Reconstructability Analysis: An Offspring of Ashby's Constraint Analysis. *Systems Research*, 3:267–271, 1986.

[16] Lihong Li, Thomas J. Walsh, and Michael L. Littman. Towards a unified theory of state abstraction for MDPs. ISAIM, 2006.

[17] Leonid Libkin. Certain Answers as Objects and Knowledge. *Artificial Intelligence*, 232:1–19, 2016. doi:10.1016/j.artint.2015.11.004.

[18] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of EMNLP*, 2023.

[19] Charles F. Manski. *Partial Identification of Probability Distributions.* Springer Series in Statistics. Springer, New York, 2003.

[20] Charles F. Manski. *Identification for Prediction and Decision.* Harvard University Press, Cambridge, Massachusetts, 2008.

[21] Judea Pearl. Causality: Models, Reasoning, and Inference. Cambridge University Press, 2009.

[22] Leonard J. Savage. The Foundations of Statistics. 1954.

[23] Glenn Shafer and Vladimir Vovk. A Tutorial on Conformal Prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.

[24] Claude Shannon. A Mathematical Theory of Communication. Bell System Technical Journal, 1948.

[25] Aviv Slobodkin, Omer Goldman, Avi Caciularu, Ido Dagan, and Shauli Ravfogel. The Curious Case of Hallucinatory (Un)answerability: Finding Truths in the Hidden States of Over-Confident Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 3607–3625, 2023. doi:10.18653/v1/2023.emnlp-main.220.

[26] Stian Soiland-Reyes, Peter Sefton, Mercè Crosas, Leyla Jael Castro, Frederik Coppens, José M. Fernández, Daniel Garijo, Björn Grüning, Marco La Rosa, Simone Leo, Eoghan Ó Carragáin, Marc Portier, Ana Trisovic, the RO-Crate Community, Paul Groth, and Carole Goble. Packaging research artefacts with RO-Crate. *Data Science*, 5(2):97–138, 2022. doi:10.3233/DS-210053.

[27] David T. Swanson. Carriers and Adequacy for Purpose: A Formal Framework for Representation-Constrained Adequacy. Manuscript, May 21, 2026.

[28] David T. Swanson. Target Stabilization and Finite Representation: A Monist Framework. Manuscript, May 21, 2026.

[29] David T. Swanson. Typed Desk-Stage Editorial Triage: Discourse vs Dialectic Entry, Contribution Type, and Auditability. Manuscript, 2026.

[30] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. 1999.

[31] Vladimir Vovk, Alex Gammerman, and Glenn Shafer. Algorithmic Learning in a Random World. Springer, 2005.

[32] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), Article 248, 2023.